The Block Universe

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by

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and

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[1] This manuscript was developed through extensive interactive discussions with OpenAI's ChatGPT, using models from the GPT-5 and GPT-6 families. ChatGPT assisted in exploring physical concepts, examining mathematical arguments, challenging assumptions, and organizing and drafting the manuscript.

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Introduction

This manuscript grew out of a personal intellectual curiosity about physics and, more specifically, about the nature of physical reality. I am not a physicist, nor do I have the formal educational background necessary to develop or propose new physical theories. My academic and professional background is in engineering, which provides some useful analytical tools, but I approach the subjects discussed here as an amateur enthusiast seeking to understand questions that I find deeply interesting.

Of particular interest to me are the *ontological* questions underlying our descriptions of the physical universe. What does it mean for a physical event to exist? What is the nature of time? What does it mean for the universe to have a complete physical history? And how should we understand our own existence as conscious observers embedded within that reality?

Modern physics provides extraordinarily successful mathematical descriptions of the natural world. Yet a mathematical framework that accurately predicts observations does not necessarily provide an intuitively satisfying picture of what physical reality *is*. The distinction between predicting what we observe and understanding what those observations imply about the nature of reality is the principal motivation for this manuscript.

My objective is to develop a coherent way of thinking about several foundational topics in physics within a common ontological framework. Rather than treating questions about spacetime, the flow of time, quantum measurement, causality, and free will as entirely separate problems, I want to explore whether they can be understood from a small number of shared principles.

The guiding approach is to adopt as few additional postulates as possible, favoring those that appear logically reasonable and compatible with established physical laws. The hope is that a relatively simple ontological perspective can provide a more unified and intellectually satisfying interpretation of phenomena that otherwise appear disconnected or paradoxical.

The Block Universe interpretation of spacetime serves as the central organizing perspective for this exploration. Beginning with the implications of relativity, the manuscript examines how a four-dimensional description of physical reality might inform our understanding of time, physical laws, quantum mechanics, conscious experience, free will, and causal consistency.

The objective is interpretive rather than predictive. I do not claim to derive new physical laws, propose a replacement for existing theories, or produce new experimentally testable predictions. Nor do I suggest that ontological reasoning alone can establish the physical truth of a particular interpretation. The aim is more modest: to explore whether a coherent interpretation of physical reality can be constructed that remains consistent with what established physics tells us, while making as few additional assumptions as reasonably possible.

Many of the ideas discussed in this manuscript have been explored extensively by physicists and philosophers. I make no claim that the underlying concepts or arguments are original. What I hope to achieve is a synthesis that is personally meaningful: a connected set of arguments that I can revisit, question, and refine as my understanding develops.

The manuscript was developed through extensive interactive discussions with ChatGPT. I used these conversations to explore ideas, challenge assumptions, construct thought experiments, examine mathematical arguments, and fill gaps in my knowledge of physics. ChatGPT also assisted substantially in organizing and drafting the text. The resulting manuscript is therefore a collaborative product of my questions, reasoning, and interpretations, together with AI-assisted explanations, analysis, and writing. This assistance should not be mistaken for independent scientific validation, and any technical claims remain subject to verification against established literature.

Throughout this exploration, I have tried to distinguish between what established physics demonstrates, what a particular interpretation suggests, and what remains speculative or unresolved. Logical self-consistency is an essential requirement, but it is not sufficient to establish physical truth. Where the proposed framework cannot explain something, I would rather acknowledge that limitation than introduce unsupported assumptions simply to preserve an appealing picture.

Ultimately, this manuscript is written primarily for myself. It is a record of an ongoing attempt to understand the universe and our place within it. Its value to me lies not in whether it contributes something new to science, but in whether it provides a coherent intellectual framework that I can return to, reconsider, and improve.

If it also proves interesting or useful to another curious reader, that would be a welcome additional outcome.

1 From Relativity to the Block Universe

Central claim of this section: Special relativity establishes that there is no observer-independent global present for spatially separated events. Minkowski spacetime provides a natural four-dimensional description of this structure. The Block Universe is a philosophical interpretation of that four-dimensional structure—not a theorem that special relativity proves by itself.

1.1 Newtonian Absolute Time and the Universal Present

In classical Newtonian mechanics, time is treated as an absolute, universal parameter: a single temporal ordering that, in principle, applies identically to all observers regardless of their location or state of motion. Within this framework, the universe can be pictured as progressing through a sequence of three-dimensional spatial states, each associated with the same universal time.

This picture naturally supports presentism, the philosophical view that only the present moment exists, while the past no longer exists and the future does not yet exist. It is important, however, to distinguish this philosophical interpretation from Newtonian mechanics itself: Newtonian physics supplies an absolute time parameter, but it does not by itself constitute a complete philosophical argument for presentism.

1.2 Einstein's Spacetime: From Separate Space and Time to Minkowski Geometry

Special relativity replaces Newton's absolute time with a spacetime structure in which spatial and temporal coordinates depend on the inertial frame used to describe events. There is no privileged inertial frame from which a universal temporal slicing of the universe can be defined.

Observers in relative motion can disagree about spatial distances and temporal durations. More importantly for the question of now, they can disagree about whether two spatially separated events are simultaneous.

Although the individual quantities Δx and Δt vary between inertial frames, the spacetime interval remains invariant:

$$\Delta s^2 = -c^2\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$$

Here c is the invariant causal speed of relativity. In ordinary units it also relates temporal and spatial coordinates: multiplying a time interval by c gives a quantity with units of length. This permits time to be represented geometrically alongside the spatial coordinates as ct, x, y, z.

The resulting Minkowski spacetime is a four-dimensional geometric structure. An object's history can be represented within it by a worldline: a continuous curve through spacetime. This geometrical description does not by itself require the additional philosophical claim that time literally “does not pass”; rather, it provides a framework in which an object's entire spacetime history can be represented as a single geometric object.

1.3 What Does “Now” Mean?

The word “now” can refer to several different things, and separating them is essential to understanding relativity.

1. **Observed now**: the events whose light signals are currently reaching an observer. For distant objects, this is affected by the finite propagation time of light.
2. **Relativistic simultaneity**: the distant events that an observer's inertial coordinate system assigns the same time coordinate as the observer's current event. These events form a hypersurface of constant time in that frame.
3. **Ontological existence**: the philosophical question of whether events in the past, present, and future exist in the same sense. Special relativity does not, by itself, settle this ontological question.

The crucial point: The relativity of “now” concerns the second meaning. Two observers in relative motion can disagree about which distant events are simultaneous with their common local event. They are not merely receiving different visual information; their inertial coordinate systems assign different simultaneity relations to spatially separated events.

1.4 The Relativity of Simultaneity

Figure 1 is a two-dimensional representation of Minkowski spacetime. A complete representation of Minkowski spacetime is four-dimensional, with three spatial dimensions and one time dimension, and therefore cannot be directly depicted in an ordinary two-dimensional diagram. For clarity, the figure suppresses two of the spatial dimensions and displays only one spatial dimension together with the time dimension. This simplification does not affect the generality of the argument: the relativistic relationships illustrated in the diagram are properties of four-dimensional Minkowski spacetime, and the conclusions drawn from them apply to the full 4D spacetime description.

Consider three inertial observers who pass through the same local event at the origin of a Minkowski diagram:

- Observer A: at rest relative to the unprimed coordinate system, with $v = 0$.
- Observer B: moving at a positive relativistic speed v' relative to A.
- Observer C: moving at a still greater speed v'', where $v'' > v'$.

For an inertial frame moving at speed v relative to A, the Lorentz transformation is

$$t' = \gamma(t - vx/c2), \qquad x' = \gamma(x - vt), \qquad \gamma = 1/\sqrt{(1-v2/c2)}$$

The key feature is the term $vx/c2$ in the transformation of time. The time coordinate assigned to a distant event depends on both its time coordinate and its spatial position in the original frame.

For an event simultaneous with the origin according to the moving observer, $t' = 0$. Therefore:

$$t' = 0 \Rightarrow t = vx/c2$$

Thus, except at x = 0, the events that B considers simultaneous with the origin do not generally satisfy t = 0 in A's frame. This is the mathematical expression of the relativity of simultaneity.

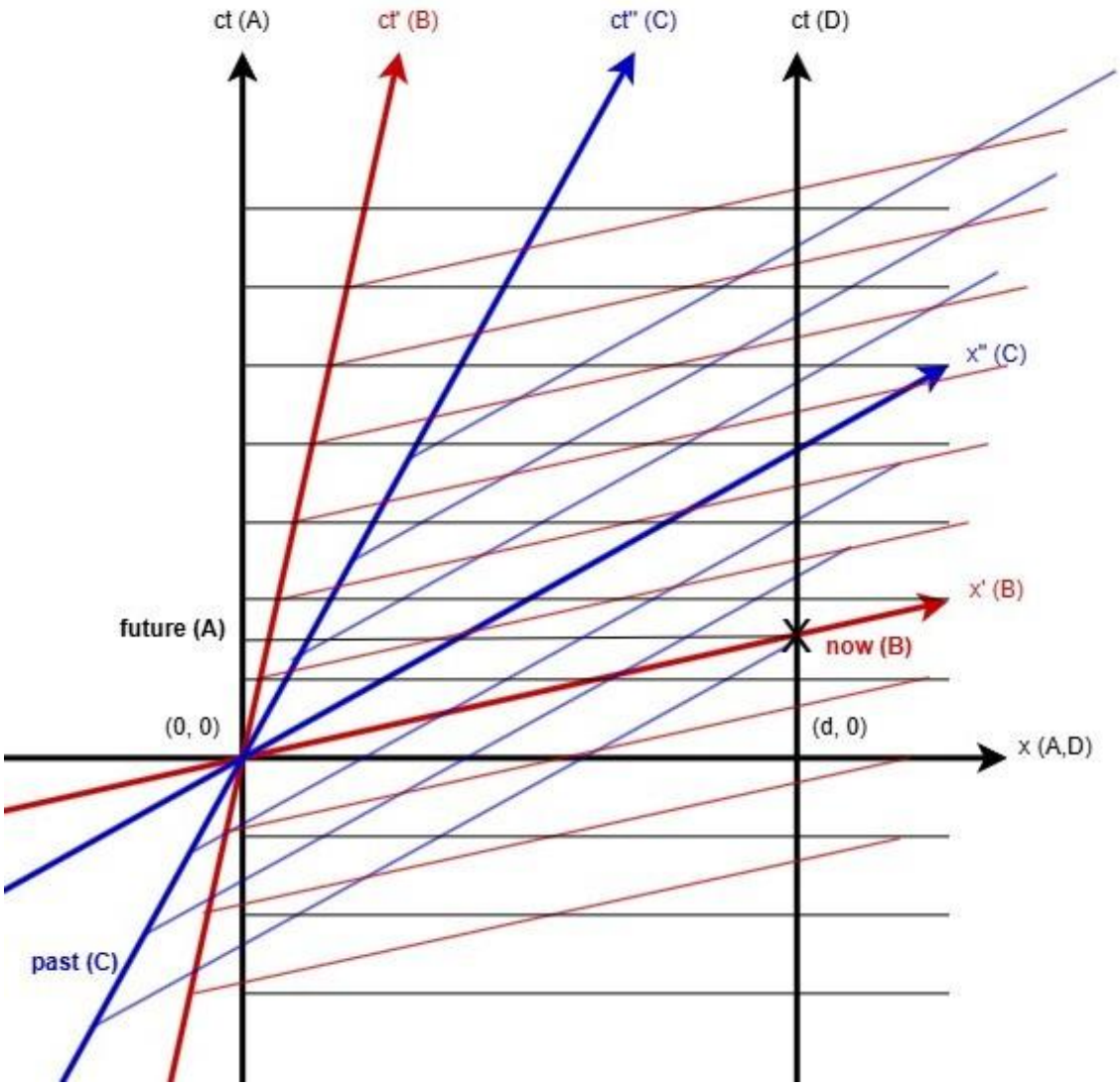


Figure 1: Minkowski spacetime diagram illustrating tilted axes and different hypersurfaces of simultaneity for inertial observers in relative motion. The vertical coordinate should be understood as ct when interpreting geometric angles.

Diagram Analysis

1. Reference-frame axes:
    - Observer A uses the unprimed x and t coordinates.
    - Observer B uses the primed x' and t' coordinates.
    - Observer C uses the double-primed x'' and t'' coordinates.
    - Observer D is local to event X (explained in later subsections)

2. Coordinate distortion:

Relative motion changes the coordinate decomposition of spacetime. In a Minkowski diagram whose vertical coordinate is ct, the time axes and simultaneity axes appear tilted relative to one another. Greater relative velocity produces a greater tilt, approaching the light-cone directions as v approaches c.

3. Hypersurfaces of simultaneity:
    - Black horizontal lines: A's constant-t surfaces.
    - Red inclined lines: B's constant-t' surfaces.
    - Blue more steeply inclined lines: C's constant-t" surfaces.

1.5 Event X: How the Same Distant Event Can Be Future, Present, or Past

Now consider a distant event X, separated from the common origin by a spatial distance $d > 0$. Choose X so that it is spacelike separated from the origin. This qualification is essential: only spacelike-separated events can have their temporal ordering or simultaneity relation changed by a change of inertial frame.

Suppose the velocity parameters are chosen so that X lies on B's hypersurface of simultaneity. Then:

1. **Observer B**: Event X satisfies $t' = 0$. B therefore assigns X the same coordinate time as the origin. In the precise relativistic sense, X is simultaneous with B's present event.
2. **Observer A**: The same event X has $t > 0$. A therefore assigns X a later time than the origin: X lies in A's future.
3. **Observer C**: For a sufficiently different relative velocity, the same event can have $t'' < 0$. C therefore assigns X an earlier time than the origin: X lies in C's past.

Nothing contradictory has happened. The three observers are using different inertial coordinate systems to describe the same two spacetime events. The disagreement arises because simultaneity for spatially separated events is not invariant.

What is—and is not—relative: For spacelike-separated events, observers can disagree about which happens first or whether they are simultaneous. For timelike - or lightlike - separated events, the causal ordering is invariant: all inertial observers agree on which event can causally precede the other.

1.6 Why There Is No Universal “Now”

The Event X example reveals the central conceptual consequence of special relativity. If A, B, and C are all inertial observers and none is physically privileged, there is no frame-independent rule that identifies one particular distant hypersurface as the present moment of the universe.

In other words, the following question has no observer-independent answer within special relativity:

“Which events happening at distant locations are occurring right now?”

A can give one answer. B can give another. C can give another. Their answers are not different because one observer has made a measurement error. They arise from the Lorentz transformation and the geometry of Minkowski spacetime.

This is the precise sense in which the “now” is relative.

Relativity of “now”: There is no single, observer-independent three-dimensional hypersurface of distant events that all inertial observers must identify as the present. Each inertial frame defines its own hypersurfaces of simultaneity.

1.7 The Ontological Question: Does Lack of Information Mean Lack of Existence?

At this point a distinction becomes essential. The finite speed of information tells us when an observer can become informed about an event; it does not by itself determine when the event exists. If event X occurs at a location spatially separated from observer B, B may be unable to receive information from X until later. That fact establishes a limit on B's knowledge, not necessarily a limit on the physical existence of X.

We therefore adopt the following working ontological principle for this argument:

Existence versus information:
If an event X is a physical event, then X does not begin to exist merely when a particular observer receives information about X. The arrival of information changes the observer's knowledge of X; it does not create X.

This can be illustrated by introducing a second observer, D, located at the same spatial coordinates as X when X occurs. D is local to X and can therefore encounter X directly. X is consequently not merely a prediction or an unreceived signal: it is an event in D's physical history.

Now consider B, who is spatially separated from X. At the moment X occurs, B may have no information about X. Nevertheless, the information generated by or causally connected with X can later reach B. When it does, B can establish retrospectively that X occurred at an earlier point in spacetime. B's earlier lack of information did not mean that X was nonexistent; it meant only that B was not yet informed of X.

1.7.1 Let the Movie Play Out

The argument can be made explicit by imagining the history unfold in sequence:

1. X occurs. Observer D, local to X, can encounter X.
2. B is not yet aware of X. Because information cannot propagate instantaneously, B may have to wait before learning about X.
3. One year passes. Information about X reaches B.
4. B becomes aware of X. B now knows that X occurred one year earlier.

5. B also knows that B existed one year earlier. Therefore, B can place X within the earlier portion of B's own physical history, rather than treating X as something that came into existence only when B learned about it.

The crucial point is that nothing about X changed when the information arrived. What changed was B's epistemic state:

$$\text{Change in B's knowledge} \neq \text{Change in the existence of X}$$

This gives the ontological argument a simple core:

X exists independently of B's receipt of information about X.

The argument does not require information to travel faster than light, nor does it claim that B can observe X before a signal from X arrives. It instead separates existence from observation. A delayed observation is still an observation of an event that occurred earlier.

1.7.2 Extending the Argument to the A–B–C Diagram

We can now return to the original diagram, in which the same spacelike-separated event X is assigned different temporal relations by three inertial observers:

A: X is in the future | B: X is now | C: X is in the past

The same ontological reasoning can be applied to each observer. The fact that information about X has not yet reached A does not imply that X is nonexistent. If A continues along its worldline, information about X can eventually reach A, at which point A can establish that X was a real physical event that lay in A's earlier future. Similarly, C can receive information from X and thereby establish that X was a real physical event in C's past.

Thus, the temporal labels future, present, and past describe the relation of X to different observers and their spacetime coordinates. They need not describe three different ontological statuses of X.

Ontological principle:
If X is a physical event, then its existence cannot depend on which observer has already received information about X. The observer-dependent distinction between past, present, and future therefore need not be an observer-dependent distinction between what exists and what does not exist.

1.8 From Relativity of “Now” to the Block Universe

We can now state the argument for the Block Universe as a sequence of distinct physical and ontological steps.

Step 1: There is no universal present.

Special relativity establishes that for spatially separated events, different inertial observers can assign different simultaneity relations. In the diagram, X is future for A, present for B, and past for C.

Step 2: Lack of information does not imply lack of existence.

An observer can fail to have information about an event simply because the information has not yet had time to arrive. The later arrival of information changes the observer's knowledge, not the earlier physical occurrence of the event.

Step 3: The same physical event therefore exists independently of the observer's information state.

Observer D, local to X, provides the direct physical case. Observers A, B, and C can subsequently acquire information about X at different stages of their respective histories. Nothing in those different information-arrival times requires X to come into or go out of existence.

Step 4: The labels past, present, and future are therefore relational.

For X, one observer can legitimately call X future, another present, and another past. If X nevertheless remains the same physical event throughout this sequence, then those labels cannot by themselves determine whether X exists.

Step 5: Past, present, and future can therefore all be physically real.

The resulting picture is not a universe in which only the observer's current three-dimensional slice exists while the other slices are nonexistent. Instead, the physical events associated with what one observer calls the past, what another calls the present, and what another calls the future can all belong to the same four-dimensional physical reality.

Relativity of simultaneity
↓
No observer-independent absolute present
↓
Existence is not created by information arrival
↓
Past, present, and future are observer-relative temporal relations
↓
All such events can be part of one four-dimensional physical reality
↓
Block Universe

In this sense, the Block Universe is the ontological interpretation developed by combining the relativistic structure of spacetime with the principle that an event's existence does not depend on when information about it reaches a particular observer.

This argument should be distinguished from the weaker claim that special relativity mathematically proves eternalism. The Lorentz transformation and Minkowski geometry provide the physical structure. The additional ontological premise—that delayed information does not create the event it informs us about—provides the bridge from that structure to the conclusion that past, present, and future events can all be physically real.

1.8.1 The Rietdijk–Putnam–Penrose Argument in This Context

The philosophical significance of relativity of simultaneity has been developed in arguments associated with Rietdijk, Putnam, and Penrose. The familiar concern is that if two legitimate inertial observers disagree about whether a distant event is present or future, while a third can assign that same event to the past, then it becomes difficult to maintain a single observer-independent boundary separating "what exists now" from "what does not yet exist."

The present argument develops that point one step further. Rather than inferring existence directly from simultaneity, it first separates information about an event from the event itself. Once that distinction is made, the fact that information reaches A, B, and C at different times does not provide a mechanism by which X itself could become real at different times for different observers.

The resulting conclusion is therefore an ontological one: if the physical event X is real for the observer local to X, and if delayed information about X does not create X, then the observer-dependent classification of X as future, present, or past does not determine its existence. The four-dimensional Block Universe supplies a natural way to represent this observer-independent collection of events.

1.9 General Relativity: Curved Spacetime and Dynamical Geometry

Special relativity is sufficient for the argument developed above for the Block Universe. The discussion of general relativity is included here for completeness, to show how the same four-dimensional spacetime framework is extended when gravity is taken into account. General relativity introduces the curvature of spacetime, in contrast to the flat spacetime of special relativity, as the geometric description of gravity. The arguments made so far, however, do not depend on spacetime being flat: they concern the four-dimensional nature of spacetime and the observer-dependence of temporal relations. Accordingly, the essential conclusions extend from flat Minkowski spacetime to curved spacetime as described by general relativity.

General relativity extends the geometric description from flat Minkowski spacetime to curved spacetime, represented mathematically by a four-dimensional manifold ($\mathscr{L}$, $g_{\mu\nu}$). The gravitational field is encoded in the spacetime metric, and the metric is dynamically related to matter and energy by the Einstein field equations:

$$G\mu\nu + \Lambda g\mu\nu = (8\pi G/c4)T\mu\nu$$

Unlike the flat geometry of special relativity, the geometry of general relativity need not be static. The Einstein field equations are dynamical equations, and particular solutions can describe evolving gravitational fields, gravitational radiation, expanding or contracting cosmological geometries, and other time-dependent phenomena.

Nevertheless, individual physical histories can still be represented within a four-dimensional spacetime geometry. Worldlines of matter and trajectories of light are curves in that geometry, while the geometry itself is determined by the gravitational field equations together with appropriate initial or boundary conditions.

Thus, general relativity preserves and deepens the geometric lesson of special relativity without requiring the claim that spacetime as a whole is universally static.

1.10 Summary: From Relativity to the Block Universe

1. Newtonian mechanics contains an absolute time parameter and therefore permits a universal notion of simultaneity.
2. Special relativity replaces absolute time with frame-dependent temporal coordinates within Minkowski spacetime.
3. For spacelike-separated events, one observer can assign an event to the future, another to the present, and another to the past. Therefore, special relativity contains no observer-independent global "now."
4. The finite speed of information limits when an observer can learn about a distant event; it does not by itself determine when that event exists.
5. If an event X is physically real for an observer local to X, the later receipt of information about X by another observer does not create X at the moment of reception.
6. Consequently, the temporal labels past, present, and future can be understood as observer-relative relations to events that belong to the same four-dimensional physical reality.
7. On this ontological interpretation, past, present, and future all physically exist, and the resulting picture is the Block Universe.

Core conclusion: Special relativity removes the possibility of a single observer-independent global "now." The finite speed of information does not rescue a distinction between "what exists" and "what has not yet been observed," because an event can exist independently of when information about it reaches a particular observer. Once that ontological principle is combined with relativity of simultaneity, the past, present, and future can be understood as different observer-relative temporal relations to events within one four-dimensional physical reality. This is the Block Universe picture.

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2 The Block Universe and Direction of the Flow of Time

2.1 From the Existence of Time to the Flow of Time

Section 1 established the Block Universe as a four-dimensional spacetime in which past, present, and future events all physically exist. The entire history of the universe can therefore be regarded as a single four-dimensional entity extending across the temporal as well as spatial dimensions. For the purposes of this discussion, we will take this block to encompass the complete history of the universe, extending back toward the Big Bang and its associated initial singularity. Every event contained within the spacetime—including events that we describe as past, present, or future—statically exists as part of the complete four-dimensional structure.

This raises a different question. If the entire spacetime exists as a whole, and time is simply one of the coordinates of that spacetime, why do we experience time as flowing in a particular direction?

The distinction between **time** and the **flow of time** is essential here. Within the Block Universe, the temporal coordinate provides an ordering of events just as spatial coordinates provide an ordering of locations. Nothing in the existence of a temporal coordinate by itself requires spacetime to physically move from one time to another. There is therefore no need to introduce a fundamental substance or process that causes the universe to advance through time.

We may therefore distinguish three related but different concepts:

- **Time as a coordinate:** the temporal dimension used to locate events within spacetime.
- **Temporal ordering:** the ordering of events along a timelike worldline, such as the sequence of physical states experienced by an observer.
- **The experienced flow of time:** the perception by a physical observer that it is progressing from one state to another, with a definite distinction between past, present, and future.

The first two are structural properties of spacetime and physical systems within it. The third is a property of an observer embedded within spacetime. The Block Universe therefore does not need to *flow* through time in order for an observer within the block to experience a flow of time.

In this sense, the flow of time is not a fundamental property of spacetime itself. It is a description of the experience of physical systems as they occupy successive states along their worldlines.

The Block Universe does not flow through time; observers experience a flow of time within the Block Universe.

2.2 The Observer as a Physical System in the Block

To understand this experienced flow, consider an observer as a physical system embedded within the four-dimensional spacetime of the Block Universe. The observer has a worldline extending through spacetime, and at different points along that worldline its physical state is different.

Let the physical state of the observer at three successive values of its temporal coordinate be represented by:

$$(T_1, S_1) \quad (T_2, S_2) \quad (T_3, S_3), \quad T_1 < T_2 < T_3.$$

All three states are part of the Block Universe. Nothing about the block itself requires us to declare that the sequence is physically flowing from S_1 to S_2 to S_3. The reverse ordering can equally be described geometrically by traversing the same worldline in the opposite coordinate direction.

At this stage, therefore, we have established temporal order but not yet the *direction of experienced time*.

The question becomes: **what physical property of the observer distinguishes one direction along its worldline as the direction of experienced time?**

2.3 The Information Structure of Successive Brain States

For a human observer, the relevant physical system is the brain. The human brain is itself an entity within the four-dimensional spacetime of the Block Universe. As the brain's worldline extends through spacetime, the physical state of the brain changes from one point to another.

Suppose that the brain is in state S_2. That state contains physical information resulting from its interaction with the world at earlier points along its worldline. The brain may contain memories of previous experiences, records of previous events, learned information, physical modifications resulting from earlier interactions, and many other forms of information that are consequences of earlier states.

Consequently, the state S_2 contains information about aspects of S_1. The later state therefore carries physical records of the earlier state.

$$S_1 \rightarrow S_2: \quad S_2 \text{ contains information about } S_1.$$

By contrast, the state S_2 does not ordinarily contain corresponding information about the future state S_3. The brain at S_2 does not ordinarily contain memories or physical records produced by events that have not yet occurred along its worldline.

$$S_2 \rightarrow S_3: \quad S_2 \text{ does not ordinarily contain information about } S_3.$$

This creates an information asymmetry between the two directions along the observer's worldline. One direction is associated with states that have already left information in the current state; the other is associated with states that have not yet done so.

This asymmetry provides a physical basis for the observer's distinction between past and future. If S_2 is the brain's present state, then S_1 is experienced as part of the **past** because information about it is

represented in the present state of the brain. Conversely, S_3 is experienced as part of the **future** because information about it is not yet represented in the brain's present state.

The experienced distinction can therefore be represented schematically as:

$$\textbf{Past} \leftarrow S_1 \quad | \quad S_2 \quad | \quad S_3 \rightarrow \textbf{Future}$$

with S_2 representing the observer's present physical state.

The important point is that this distinction does not require the future to be nonexistent. S_3 already exists as part of the Block Universe. What distinguishes it from S_1 from the perspective of the brain is not its ontological status, but the information contained in the brain's current physical state.

Past, present, and future are therefore not necessarily three different modes of existence within the Block Universe. They are three different relationships between an observer's present physical state and other regions of spacetime.

2.4 The Direction of Experienced Time

We can now identify what is meant by the *direction* of the flow of time for an observer.

The brain does not merely occupy a sequence of physical states. Its successive states contain records of events associated with preceding states. The direction in which these records accumulate establishes an asymmetry along the brain's worldline.

At S_2, the brain contains information about S_1, while it does not ordinarily contain information about S_3. The brain therefore experiences S_1 as past and S_3 as future.

In this sense, the direction of experienced time can be associated with the direction of the observer's worldline along which physical records of earlier states are established in later states.

It is important, however, not to equate this simply with the statement that the brain must monotonically accumulate information. A brain can forget information, lose memories, or undergo local decreases in stored information. The relevant property is more fundamental: **the causal-information structure of successive physical states is asymmetric.** Later states ordinarily contain records and consequences of earlier states, while earlier states do not contain corresponding records produced by later states.

The direction of experienced time is therefore not defined by the absolute quantity of information stored in the brain. Rather, it is defined by the asymmetric orientation of information and physical records along the observer's worldline.

2.5 The Thermodynamic and Gravitational Origin of the Information Asymmetry

The existence of an information asymmetry in the brain raises a further question. Why should physical records be oriented in this particular temporal direction at all? Why should later physical states contain records of earlier states rather than the reverse?

This is where the thermodynamic history of the universe enters the argument. The universe possesses a profound thermodynamic asymmetry. The early universe was in an extraordinarily low-entropy state, while entropy has generally increased through the subsequent evolution of the universe. This provides an intrinsic physical asymmetry within the Block Universe itself.

However, the low-entropy condition is not the whole physical story. The early universe was extremely homogeneous and isotropic on large scales, but it was not perfectly homogeneous. Small primordial density fluctuations were present. These fluctuations provided the seeds for gravitational structure formation: under the influence of gravity, initially small differences in density could grow, eventually producing galaxies, stars, planets, and the complex physical environments in which chemistry, biology, information processing, and observers could arise.

This point is especially important because gravity changes the way we think about entropy. A highly smooth matter distribution can correspond to very low gravitational entropy, whereas gravitationally clumped structures—most dramatically, black holes—represent states of much greater gravitational entropy. The early universe therefore combined an exceptionally smooth large-scale matter distribution with small fluctuations capable of growing into structure. The subsequent formation of gravitational structure is part of the physical history through which the universe developed from its unusually low-entropy initial condition into the complex, high-entropy states found later in the Block.

The primordial density fluctuations should not, however, be identified as the fundamental source of the direction of experienced time. Rather, they are an important physical ingredient in the mechanism by which the thermodynamic asymmetry of the early universe becomes expressed through the formation of structure and irreversible processes. The low-entropy boundary condition supplies the fundamental asymmetry, while the initial fluctuations allow gravitational evolution to generate the structured physical systems in which that asymmetry can be manifested.

The important point is that the thermodynamic asymmetry should not simply be identified with the direction of experienced time. Saying that entropy increases toward one end of the block does not, by itself, explain why that direction should be experienced by an observer as the future rather than the past.

Instead, the thermodynamic and gravitational asymmetries provide the physical conditions under which temporally asymmetric processes and records can arise. As structure forms and physical systems interact, information about preceding states becomes encoded in subsequent states through processes such as measurement, interaction, memory formation, and the formation of physical records. These processes are thermodynamically irreversible at the macroscopic level and therefore contribute to the asymmetric distribution of physical information across successive states.

Thus, the relevant chain of explanation is not simply:

Low entropy → flow of time.

Nor is it necessary to treat primordial inhomogeneity as an independent arrow of time. A more complete physical chain is:

low-entropy initial condition → primordial density fluctuations → gravitational structure formation → thermodynamically irreversible processes → asymmetric formation of physical records → information asymmetry in observers → experienced direction of time.

In this formulation, entropy does not define the direction of experienced time. Rather, the low-entropy initial condition establishes the fundamental thermodynamic asymmetry of the universe, while primordial fluctuations provide the seeds through which gravitational structure and increasingly complex physical systems develop. The resulting thermodynamically asymmetric processes produce information-bearing records in a preferred temporal orientation. Human observers, as physical information-processing systems embedded within this history, inherit that information asymmetry, which in turn provides the physical basis for their experience of a directed **flow of time**.

2.6 Why Local Entropy Decreases Do Not Reverse the Flow of Time

This distinction also resolves an apparent difficulty with identifying the direction of time directly with entropy.

Entropy is not required to increase monotonically in every local region of spacetime. Local entropy can decrease, provided that this decrease is compensated by a sufficient increase elsewhere so that the total entropy of the appropriate closed system increases. Consequently, there can be local fluctuations and processes in which entropy decreases even while the overall thermodynamic arrow remains unchanged.

If the direction of experienced time were simply defined as the direction of increasing local entropy, an observer might appear to experience a reversal of time whenever it entered a region in which local entropy happened to decrease. This is not what we observe.

Our experience of time therefore cannot be identified directly with the local entropy gradient.

Instead, the relevant connection is between the thermodynamic asymmetry of the physical environment and the asymmetric formation of records within physical systems. A local fluctuation in entropy does not, by itself, reverse the causal and informational structure of an observer's physical states. The observer continues to contain records of what we call the past and not of what we call the future.

Thus, the direction of experienced time is a property of the observer's information-bearing physical state, while the thermodynamic arrow provides the broader physical asymmetry from which this information structure emerges.

2.7 Past, Present, and Future as Relations Within the Block

We can now return to the central premise of the Block Universe.

At the level of the complete four-dimensional spacetime, S_1, S_2, and S_3 all exist. Nothing has been created when the brain reaches S_2, and nothing ceases to exist when the brain passes beyond it. The entire sequence is already part of the four-dimensional structure.

Nevertheless, the brain at S_2 has a particular informational relationship to the rest of spacetime. It contains records of states associated with S_1, while it does not yet contain corresponding records of S_3. It is this asymmetric informational relationship that gives rise to the observer's distinction between past and future.

The familiar human experience of temporal passage can therefore be understood without requiring time itself to flow through the Block Universe.

The past is not a region of spacetime that has ceased to exist. The future is not a region of spacetime that has yet to come into existence. Both are already present in the block. What changes from one point along the observer's worldline to another is the physical state of the observer and the information encoded within that state.

Conclusion: *The Block Universe is static with respect to its complete four-dimensional existence, but it is not temporally symmetric in its physical contents. The low-entropy condition of the early universe establishes a thermodynamic asymmetry within the block. That asymmetry gives rise to a corresponding asymmetry in the formation of physical records. Human brains, as physical information-processing systems embedded within spacetime, inherit this asymmetry: later brain states contain records of earlier states, while earlier states do not contain records of later states. The brain consequently experiences the direction of this information asymmetry as the flow of time, distinguishing past, present, and future.* ***Thus, the flow of time is not a fundamental motion of the Block Universe itself, but an emergent feature of observers physically embedded within its asymmetric structure.***

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3 The Laws of Physics in a Static Block Universe

3.1 How to Think About the Laws of Physics in a Static Block Universe

Because human observers perceive time as flowing, we naturally tend to think of the laws of physics in dynamic terms. We imagine the universe as having a particular physical state at the present moment, with the laws of physics determining what happens next. In this picture, the current "now" can be thought of conceptually as an input, the next "now" as an output, and the laws of physics as the rules that relate the two:

$$\text{Current state} \rightarrow \text{laws of physics} \rightarrow \text{next state}$$

This is a useful conceptual model of physical evolution. It corresponds naturally to our experience of time: one state exists "now," physical processes occur, and another state subsequently becomes the new "now.", similar to an algorithm with inputs and outputs.

This algorithm analogy is intended only to make this conceptual distinction clear. It should not be interpreted literally. The laws of physics need not constitute an algorithm in the computational sense, nor do they necessarily imply that the universe is deterministic. An algorithm, in the conceptual sense used here, could contain a probabilistic or random component. The important point is simply that the dynamic description treats the physical state at one time as being related by the laws of physics to the physical state at another time.

The Block Universe provides another way of describing the same physical reality.

If the past, present, and future all exist as parts of a complete four-dimensional spacetime, then the universe does not need to be thought of as continually generating a new "now." There is no physical process in which the present moment disappears and is replaced by a newly created future moment. The complete history of the universe is already represented within the four-dimensional structure.

This changes the way in which we can think about the laws of physics.

Rather than regarding the laws as rules that are continually being executed as the universe advances from one moment to the next, we can regard them as being **enshrined in the self-consistent physical relationships contained within the Block Universe itself**.

For example, instead of imagining a physical object as repeatedly being updated from one position to another,

$$x(T_1) \rightarrow x(T_2) \rightarrow x(T_3)$$

we can regard its complete worldline as part of the four-dimensional physical structure. The laws of physics describe the relationships between the different points along that worldline and between that worldline and the other physical systems with which it interacts.

From this perspective, the laws of physics do not have to be thought of as something external to the Block Universe that is continually acting upon it. Rather, the physical contents of the block exhibit the relationships described by those laws.

Dynamic view:
The laws relate the current state to subsequent states.

Block Universe view:
The complete physical history exhibits the relationships described by the laws.

These are not necessarily competing descriptions. They are two ways of conceptualizing the same physical relationships.

The dynamic description emphasizes **evolution**. The Block Universe description emphasizes **self-consistency**.

This distinction also changes how we should think about our own experience of discovering the laws of physics. Because we experience time as flowing, we encounter the physical world one state at a time. We observe an object in one state, then observe it in another, and discover that the relationship between those states is consistent. We perform an experiment, obtain a result, repeat the experiment, and find the same physical relationship again.

In this sense, our perception of the flow of time causes us to encounter the self-consistent structure of the Block Universe sequentially. What we describe as a law of physics is, from this perspective, a relationship that remains consistently manifested throughout the relevant regions of spacetime.

We therefore do not need to imagine that the universe is repeatedly "applying" the laws of physics in real time in order for physical laws to govern what we observe. The complete spacetime structure can simply embody those relationships. As observers embedded within that structure, we encounter them sequentially and interpret their persistent consistency as the laws governing the evolution of physical systems.

This provides an important conceptual bridge between the static and dynamic descriptions of reality.

The Block Universe is static as a four-dimensional entity, while the physical relationships within it are precisely those that, when experienced along an observer's worldline, appear as dynamical evolution.

This distinction will become particularly important when considering quantum mechanics. If the laws of physics can be understood either as governing successive physical states or as describing the self-consistent relationships contained within the complete spacetime structure, then the apparent transition from one quantum state to another must also be examined from both perspectives. In particular, it raises

the question of what, within a static four-dimensional description, the conventional notion of **wave-function collapse** is actually describing.

4 The Block Universe and Quantum Mechanics

4.1 Relativity, the Block Universe, and the Measurement Problem

The Block Universe developed in the preceding sections is based on the four-dimensional spacetime description provided by the theory of relativity. Relativity, and the Block Universe interpretation built upon it, are normally regarded as *classical* in the sense that the spacetime geometry itself is not treated as a quantum object. In particular, general relativity does not require spacetime, or the gravitational field represented by its geometry, to be quantized.

For the purposes of the argument developed in this section, however, that distinction is not central. We are not attempting to construct a deeper theory that unifies general relativity with quantum mechanics, nor are we attempting to determine the ultimate theory of quantum gravity. The fact that our present theories are expected to be incomplete at certain physical regimes does not prevent us from asking what each theory says within the domains in which it successfully applies.

Both relativity and quantum mechanics are, in important respects, understood as parts of a deeper physical description. General relativity encounters regimes in which its classical description of spacetime is expected to become inadequate, while quantum mechanics and quantum field theory do not by themselves provide a complete description of gravitation and spacetime at the most extreme scales. The existence of these limitations is an important motivation for fundamental physics, but it is not the problem we are addressing here.

Instead, the question is considerably more specific. Quantum mechanics appears to contain a peculiar transition between two apparently different kinds of evolution. According to the Schrödinger equation, a quantum state evolves continuously and deterministically according to a well-defined mathematical rule. Yet when a measurement is made, the theory is conventionally described as undergoing a sudden *collapse of the wave function* to one of the possible measurement outcomes.

This raises the familiar **measurement problem**: if the quantum state evolves according to the Schrödinger equation, and if the measuring apparatus is itself a physical system governed by quantum mechanics, what physical process causes the apparently discontinuous transition from a superposition of possible outcomes to a single observed outcome?

The problem is therefore not, at least for the purposes of this discussion, whether quantum mechanics is the final theory of nature. Nor is it necessary to solve the problem of quantum gravity. The question is whether the four-dimensional perspective developed in the preceding sections can provide a different way of understanding what is described as wave-function collapse.

In other words, we will temporarily accept the successful mathematical framework of quantum mechanics and ask whether the apparent mystery of measurement is partly a consequence of the way we ordinarily describe events as unfolding in time. The Block Universe gives us a very different starting point: rather than imagining reality as something that is continuously being created from moment to moment, we begin with the complete four-dimensional spacetime structure. The question then becomes

whether quantum measurement needs to be understood as a physical event in which the universe literally selects and creates one outcome, or whether the apparent collapse can be understood differently within the complete spacetime description.

$$\text{Unitary quantum evolution:} \quad i\hbar\, \partial|\psi\rangle/\partial t = \hat{H}|\psi\rangle$$

$$\text{Measurement:} \quad |\psi\rangle \rightarrow |\text{outcome}\rangle$$

The second transition is the feature we will examine. The aim is not to dismiss quantum mechanics, nor to replace its experimentally successful predictions with an unsupported alternative, but to investigate whether the Block Universe changes the conceptual interpretation of this transition.

4.2 Entanglement and Superposition Are Distinct Concepts

Before continuing with the discussion of the measurement problem, it is important to distinguish two concepts that are often treated as though they were interchangeable: **superposition** and **entanglement**. They are related in quantum mechanics, but they are not the same thing.

A system can be in a superposition without being entangled with another system. For example, a single quantum system may be described by

$$|\psi\rangle = (1/\sqrt{2})(|R\rangle + |B\rangle).$$

Here the quantum state is a superposition of two possible states. Nothing about this expression requires the existence of a second system.

Entanglement is instead a property of a *joint quantum state*. Two systems are entangled when their joint state cannot be written as the product of an independent state for each subsystem. For example, the state

$$|\Psi\rangle = (1/\sqrt{2})(|R\rangle_A|B\rangle_B + |B\rangle_A|R\rangle_B)$$

cannot be decomposed into a state for A multiplied by an independent state for B. The physical description therefore belongs to the pair as a whole rather than being reducible to two independent quantum states.

This should also be distinguished from ordinary classical correlation. Two colored balls can be correlated—for example, one is red and the other blue—without either ball being in a quantum superposition and without the pair being quantum mechanically entangled. In such a case the balls possess individual physical states, and the correlation simply relates those states.

$$\text{Correlation} \neq \text{Superposition} \qquad \text{Entanglement} \neq \text{Superposition}$$

Quantum entanglement can involve superposition, but the defining feature of entanglement is the non-separability of the joint quantum state. This distinction becomes important when interpreting Bell

experiments. Bell's theorem is not simply a statement that quantum systems are in superposition. It concerns whether the observed correlations of entangled systems can be reproduced by a suitable class of underlying local hidden-variable descriptions.

4.3 Bell's Theorem and the Assumptions Behind Bell Inequalities

Bell's theorem addresses a precise question: can the correlations predicted by quantum mechanics, and observed experimentally, be reproduced by a theory in which measurement outcomes arise from underlying physical variables while respecting locality and certain statistical assumptions?

Let λ denote the relevant underlying physical state of a pair of systems. Alice chooses a measurement setting a and obtains an outcome A; Bob chooses a measurement setting b and obtains an outcome B. In a conventional local hidden-variable description, the outcomes are determined probabilistically or deterministically by the local setting and the underlying state. In the deterministic special case this may be written schematically as

$$A = A(a, \lambda), \quad B = B(b, \lambda).$$

The locality condition relevant to Bell's theorem is more precise than the ordinary statement that faster-than-light communication is impossible. Bell locality requires that, once the relevant underlying physical state λ is specified, the joint probability of the outcomes factorizes into independent local response probabilities:

$$P(A,B \mid a,b,\lambda) = P(A \mid a,\lambda)\, P(B \mid b,\lambda).$$

In this formulation, Alice's local outcome probabilities depend on her setting and λ, while Bob's depend on his setting and λ. The distant measurement setting or outcome does not provide additional information about the local outcome once these variables have been specified.

This condition should be distinguished from operational no-signaling, which requires that the observable probability distribution of Alice's outcomes cannot be changed by Bob's choice of measurement setting, and vice versa. No-signaling is a weaker condition than Bell locality. Quantum mechanics respects operational no-signaling even though its entangled-state correlations cannot be reproduced by Bell-local, measurement-independent hidden-variable models.

This distinction is important for the present argument. Our objective is not merely to preserve the impossibility of faster-than-light communication, but to examine whether a local hidden-variable description can reproduce quantum correlations when measurement independence is relaxed.

Bell inequalities require more than Bell locality alone. A central additional assumption is measurement independence, also called statistical independence:

$$P(\lambda \mid a,b) = P(\lambda).$$

This expresses the assumption that the underlying physical state λ is statistically independent of the measurement settings selected by Alice and Bob. It is often described informally in terms of "free choice," but the mathematical content is statistical rather than metaphysical. The assumption is not a consequence of special relativity.

Bell's original 1964 argument treated some of these assumptions implicitly, while later formulations made the statistical assumptions more explicit. For present purposes, the important logical point is that the Bell inequality follows only for the class of theories satisfying the relevant Bell assumptions. Consequently, an experimental violation establishes that the observed correlations cannot be reproduced by a theory satisfying all of those assumptions simultaneously.

$$\text{Bell inequality violated} \Rightarrow \text{at least one required assumption fails.}$$

This statement is weaker, and more precise, than the frequently encountered claim that Bell's theorem proves that physical influences must travel faster than light. Bell-inequality violations establish correlations that cannot be reproduced by Bell-local, measurement-independent hidden-variable models. They do not by themselves identify which assumption must be abandoned, nor do they establish that a usable faster-than-light signal has been transmitted.

4.4 The Experimental Result: What the Bell Correlations Actually Tell Us

The experimental result should first be taken at face value. Quantum experiments produce correlations between spatially separated measurement outcomes that violate Bell inequalities. Whatever interpretation is ultimately adopted, the observed correlations are an empirical fact that any adequate physical account must reproduce.

The usual conceptual question is then framed as follows: if Alice and Bob are separated by a spacelike interval, how can their outcomes be correlated more strongly than permitted by a local hidden-variable model?

A common response is to attribute the correlations to a nonlocal feature of the quantum state, sometimes described informally as "spooky action at a distance." But the experimental result itself is more limited. It tells us that the correlations cannot be explained by a particular class of separable, Bell-local hidden-variable models satisfying measurement independence. It does not by itself establish that one measurement physically changes the distant system instantaneously.

This distinction motivates the alternative developed here. Rather than beginning by asking what physical influence travels from Alice to Bob, we ask whether the observed correlation could instead be understood as a relationship among events already contained within a complete four-dimensional physical history.

4.5 The Block Universe and Measurement Independence

The Block Universe provides a natural perspective from which to question measurement independence as a fundamental requirement. In the four-dimensional description developed earlier, the complete physical history contains events at different spacetime locations and times. The fact that one event occurs later along a particular worldline does not mean that it must be statistically independent of every earlier physical variable.

Suppose λ describes an underlying physical property of a quantum system and a and b describe measurement settings that occur later in the experimental history. The conventional statistical-independence assumption requires

$$P(\lambda \mid a,b) = P(\lambda).$$

The Block Universe proposal considered here instead allows

$$P(\lambda \mid a,b) \neq P(\lambda).$$

It is important to distinguish what the Block Universe permits from what it logically requires. The existence of a complete four-dimensional physical history does not, by itself, imply that measurement independence must be violated. A Block Universe can contain statistically independent variables just as it can contain correlated ones. The proposal developed here is therefore not that measurement dependence follows necessarily from the Block Universe, but that the Block Universe provides an ontological framework in which such dependence can be considered without automatically invoking a physical influence propagating between distant events.

The meaning of this is important. It does not mean that Alice's or Bob's later measurement sends a physical influence backward in time to alter λ. Nor does it require us to postulate a new force or signal. Rather, the proposal is that the underlying physical variables and the measurement settings may possess statistical correlations as part of the complete four-dimensional history.

This is the central conceptual move of the present argument:

Correlation across spacetime ≠ interaction across spacetime.

In an ordinary dynamical description, one naturally asks what event causes another event to occur. In a Block Universe description, relationships between events can instead be considered as properties of the complete spacetime structure. The absence of a dynamical influence from A to B therefore does not imply the absence of a physical correlation between A and B.

However, this ontological possibility should not be confused with a complete physical explanation. Allowing measurement dependence does not explain why the correlations take precisely the form predicted by quantum mechanics. A fully developed physical theory would need to specify the relevant

probability distributions and reproduce the experimentally observed correlations, including their no-signaling properties.

The present proposal makes a narrower claim. It questions whether measurement independence must be regarded as a fundamental physical principle and explores the consequences of relaxing it within a globally consistent Block Universe. Once measurement independence is relaxed, the conventional Bell inequality no longer constrains every local hidden-variable description in the same way.

The Block Universe thus provides a possible ontological interpretation of measurement-dependent correlations, without claiming that the Block Universe alone derives those correlations or establishes that measurement independence is actually violated in nature.

4.6 Ontically Random Hidden Variables in the Block Universe

The hidden variables considered here are not intended to restore a deterministic theory beneath quantum mechanics. This distinction is essential. The proposal retains the possibility that quantum randomness is **ontic**: the underlying physical variable itself can be genuinely random rather than merely unknown because of incomplete information.

Consider a simplified double-slit experiment. Introduce an underlying variable λ with a hundred possible values. Values 1–10 might correspond to histories in which the experimental record indicates the photon went through the left slit; values 11–20 might correspond to histories indicating the right slit; and values 21–100 might correspond simply to histories in which an interference pattern is ultimately observed. The last category makes no claim about a particular classical trajectory through both slits. It is merely a label for the observed interference outcome.

The particular value of λ is taken to be genuinely random. Before the relevant event is encountered along an observer's worldline, the observer cannot know which value will be realized. The point is not that the value was secretly determined by some deeper deterministic variable.

What is added is a correlation between λ and the measurement history. If the experimental arrangement includes a measurement of which slit the photon traverses, the value of λ encountered by the observer is correlated with that measurement arrangement. When the measurement records the left slit, the observer discovers a value in the corresponding group. The measurement does not cause the photon to decide, at that instant, which history it had followed. Rather, the measurement event and the photon history are components of one self-consistent Block Universe history.

The same conceptual structure can be extended to Bell experiments. Let λ be an ontically random hidden variable associated with an entangled pair, while a and b are Alice's and Bob's measurement settings. The complete history may contain correlations among

$$\lambda, \quad a, \quad b, \quad A, \quad B.$$

Those correlations do not require a signal to propagate from Alice to Bob, nor from Bob to Alice. They are simply correlations among physical events contained in the same four-dimensional history. Quantum mechanics supplies the experimentally successful probability structure that those correlations must reproduce, while the hidden variable identifies which particular physical history is encountered along an observer's worldline.

In this picture, quantum mechanics need not be replaced by a deeper deterministic theory. The proposal instead retains the quantum probability distribution and adds an ontological interpretation of the individual outcomes. The hidden variable answers the question, “Which particular outcome occurs in this physical history?” Quantum mechanics answers the statistical question, “With what probabilities are the different outcomes realized?”

Quantum mechanics → statistical structure of possible histories

Block Universe + hidden variables → particular realized history

This is therefore distinct from a proposal in which quantum probabilities are merely epistemic and a deeper deterministic theory is assumed to exist. Such a theory might, in principle, resemble the relationship between microscopic dynamics and macroscopic statistical laws, or between detailed physical dynamics and probabilistic weather forecasts. That is a much stronger hypothesis and is not required for the present argument.

4.7 Measurement as a State Transition Along a Worldline

The Block Universe perspective also changes the conceptual status of measurement itself. In the ordinary dynamical picture, a measurement is often represented as a sudden transition of the quantum state:

$$|\psi\rangle \rightarrow |\text{outcome}\rangle.$$

This notation naturally suggests that the universe was previously in a superposition and that, at the instant of measurement, some physical process caused one possibility to become real while the others ceased to be real. This is the source of much of the mystery associated with wave-function collapse.

In the Block Universe picture developed here, the measurement can instead be understood as a physical correlation between the measured system, the measuring apparatus, and the observer. The observer is itself a physical system with a worldline. Successive states of that observer may be represented schematically as

$$S_1 \rightarrow S_2 \rightarrow S_3.$$

At S_2, the observer has a physical record of a measurement outcome. The observer therefore experiences the measurement as an event occurring at a particular time. But from the four-dimensional

perspective, there is no requirement that the entire universe undergo a sudden, globally propagating collapse at S_2. The measurement is one part of the complete spacetime structure.

This does not deny that a quantum measurement produces a physical change in the measuring apparatus or observer. It denies only that this local physical change must be accompanied by an instantaneous change propagating through the entire universal quantum state.

The proposal can therefore be summarized as follows:

Quantum state and its evolution → statistical constraints on physical histories

Ontically random hidden variable → particular measurement outcome

Block Universe → complete four-dimensional consistency of the measurement history

Observer's worldline → sequential encounter with the physical states and records constituting the measurement

On this interpretation, wave-function collapse is not a fundamental physical process occurring throughout spacetime. What is conventionally described as collapse may instead be a feature of the observer's state transition and the information available along that observer's worldline.

4.8 The Scope and Test of the Proposal

The argument developed so far should be stated modestly. It does not claim that Bell's experiments are mistaken, that quantum mechanics is empirically incorrect, or that a complete alternative theory has already been constructed. The experimental correlations remain exactly what they are: correlations that violate Bell inequalities.

The proposal instead questions the interpretation of those violations. Bell's result shows that the observed correlations cannot be reproduced by hidden-variable theories satisfying all of the assumptions required to derive Bell's inequality. The present proposal retains locality and quantum randomness but rejects measurement independence, allowing the hidden variables to be statistically correlated with the measurement settings as part of the complete Block Universe history.

Locality — retained
Quantum probability distributions — retained
Ontic randomness — retained
Hidden variables — introduced to identify individual outcomes
Measurement independence — rejected
Instantaneous wave-function collapse — unnecessary
Superluminal interaction — unnecessary

The crucial mathematical test is whether this framework can reproduce the precise correlations observed in Bell experiments while maintaining the stated locality condition. The CHSH formulation provides the cleanest way to make that test explicit. The derivation of the CHSH bound, the exact role of measurement independence, and an explicit measurement-dependent local hidden-variable construction are presented in Appendix A. The main text will use those results rather than interrupt the conceptual development with the full algebra.

4.9 Delayed-choice and Quantum Eraser Experiments

In these experiments, quantum systems are prepared so that which-path information is correlated with another degree of freedom or an entangled partner. A subsequent measurement can be arranged so that the resulting data distinguish paths or reveal interference within appropriately selected subsets.

In some arrangements, the relevant measurement occurs after the signal particle has already been detected. This can create the impression that the later measurement somehow changes whether the earlier particle exhibited wave-like or particle-like behavior. However, the experimentally relevant result is a correlation between measurement records. The unconditioned distribution of earlier detection events does not change according to the later measurement choice. Interference is revealed in particular subsets of the data when the corresponding measurement records are compared.

The Block Universe provides a natural interpretation. All the detection events, measurement settings, and correlations are already part of the complete four-dimensional history. Comparing the records reveals relationships among those events; it does not create or modify the events themselves. No physical signal needs to travel backward through time, and no earlier measurement record needs to be rewritten.

The Block Universe framework developed in this section also clarifies the role of wave-function collapse in these experiments. Quantum mechanics uses the wave function to describe the statistical structure of possible measurement outcomes. When an outcome is obtained, the mathematical formalism updates the quantum state accordingly. This update need not be interpreted as a physical process propagating instantaneously through spacetime. In the proposed Block Universe account, the actual physical history already contains the particular measurement settings, interactions, and outcomes that occurred. The mathematical collapse reflects conditioning the quantum description on an observed result, rather than a physical operation that creates the corresponding history. The delayed-choice aspect therefore introduces no additional physical mystery requiring a special collapse mechanism.

The same conceptual framework applies to the Bell experiments discussed earlier. In both cases, the actual measurement settings and outcomes are components of one complete spacetime history. Our proposal allows correlations between hidden variables and measurement settings, rather than assuming measurement independence as a fundamental requirement.

The delayed-choice and quantum eraser experiments do not independently establish that measurement independence is violated. Their observed correlations can be calculated using standard quantum mechanics. Their significance for the present argument is conceptual: they illustrate how apparently

paradoxical relationships between temporally separated events can be understood without requiring physical influences to propagate backward in time.

The Block Universe does not change the experimental predictions. It changes the proposed ontology of the events to which those predictions refer.

Delayed-choice and quantum eraser experiments appear mysterious when physical reality is imagined as a sequence of states that must be progressively created, with later measurements somehow determining the status of earlier events. Within the Block Universe, this difficulty is substantially reduced. The complete physical history contains the actual measurement settings and outcomes. Their correlations are properties of that history, while the quantum formalism describes their statistical relationships. There is no need to introduce a physical mechanism by which a later choice alters an earlier event, or by which mathematical wave-function collapse propagates through spacetime. The apparent paradox arises not from the observed correlations themselves, but from the additional demand that those correlations must be produced by a dynamical influence between the events.

4.10 The Core Result of Section 4

We can now state the principal result of the argument developed in this section. The detailed CHSH derivation and the explicit construction are given in Appendix A; here we summarize what that analysis establishes.

Bell's theorem does not establish that quantum correlations are impossible for every local hidden-variable theory. It establishes that they are impossible for the narrower class of local hidden-variable theories that also satisfy measurement independence and the other assumptions required by the Bell inequality.

Once measurement independence is relaxed, an explicit local-response model can reproduce the exact singlet probabilities:

$$P(A=a,B=b|x,y) = [1 - ab(x\cdot y)]/4.$$

The model does this through a setting-dependent hidden-variable distribution:

$$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(x\cdot y)]/4.$$

The local response functions remain

$$A=\lambda_1, \quad B=\lambda_2.$$

Therefore, the observed Bell correlation need not be represented mathematically as a direct dependence of Alice's outcome on Bob's setting or vice versa.

4.11 What Has Been Shown—and What Has Not

It is useful to separate the conclusions that follow from the mathematics from the larger interpretation being proposed. The technical derivation supporting these conclusions is collected in Appendix A.

Shown by the construction:

- Measurement-independent local hidden-variable models satisfy the CHSH bound.
- Quantum mechanics predicts, and experiments observe, violations of that bound.
- The CHSH derivation uses a common hidden-variable distribution across the different setting combinations.
- Relaxing measurement independence removes that particular step.
- An explicit measurement-dependent model with local response functions reproduces the singlet joint probabilities exactly.
- The model preserves the usual local marginal statistics and therefore does not create an operational signaling channel.

Not yet shown:

- that the Block Universe is the physical mechanism responsible for measurement dependence;
- that the same construction can reproduce the full quantum formalism;
- that a complete relativistic theory can generate the required joint distributions;
- that the resulting interpretation resolves every aspect of the measurement problem;
- or that the concept of free will is thereby eliminated or preserved in any particular philosophical sense.

The final point is especially important. The mathematics has brought us to the boundary between quantum foundations and a much older philosophical question: if a complete four-dimensional physical history contains the future as well as the past, in what sense are the choices made by observers genuinely free?

4.12 The Hidden Variable as Part of a Complete History

The complete history can now be represented schematically, building on the technical construction developed in Appendix A:

$H = \{\text{spacetime geometry, preparation, } \lambda\text{, settings, outcomes, records, observer states}\}$.

From the perspective of an observer moving along a worldline, these components are encountered sequentially. From the Block Universe perspective, they form one complete four-dimensional physical structure.

The hidden variable therefore does not need to be imagined as a little object carrying a secret message from the source to the measurement devices. It is better understood, at this stage, as part of the specification of the complete physical history.

The key unresolved issue is then no longer simply how the hidden variable travels from one place to another. The deeper question becomes why the complete history has the correlations that it does.

4.13 The Natural Transition to Section 5

At the beginning of this section, the Block Universe was introduced as a way of rethinking quantum measurement. Bell's theorem then forced the discussion to become more precise. The detailed CHSH analysis, including the exact role of measurement independence, has been set out separately in Appendix A so that the main text can remain focused on the conceptual argument.

Once measurement independence is relaxed, the explicit construction in Appendix A shows that the quantum singlet correlations can be reproduced while retaining local response functions. The Block Universe provides a conceptual setting in which the resulting correlations can be interpreted as relationships within a complete four-dimensional history rather than as superluminal interactions.

But this immediately raises the question that cannot be avoided:

If the complete physical history contains correlations between hidden variables and future measurement settings, what does this imply about the freedom of the observers who make those choices?

That question must be treated separately from the mathematical Bell argument. Measurement independence is a statistical assumption. Human freedom is a philosophical and physical concept involving agency, deliberation, causation, information, and the structure of an observer's worldline. The two questions are related, but they are not identical.

The next section will therefore take the Block Universe itself as the starting point and examine the apparent tension between a complete four-dimensional history and the ordinary concept of **free will**.

5 The Block Universe and Free Will

What does a four-dimensional universe imply about human choice?

5.1 The question must begin with a definition of free will

The question of whether the Block Universe is compatible with free will cannot be answered without specifying what free will means. The term is used for several different claims: that people deliberate and act on reasons; that their actions can be attributed to them; that alternative outcomes may be physically possible; or, more strongly, that an agent possesses a special power to select which outcome becomes actual.

These claims should not be conflated. In particular, the existence of deliberation does not by itself establish a freedom that stands apart from the physical history. To keep the distinction visible, we will use three terms:

Deliberative will refers to the real physical processes through which a conscious observer evaluates alternatives, drawing on conscious and unconscious information processing, and arrives at a choice. The term does not imply any exemption from physical laws.

Perceived free will refers to the observer's subjective experience of having controlled which outcome occurred. The experience itself is real; whether its implied power of selection exists is a separate question.

Fundamental free will, in the strong sense examined here, refers to an additional power of an agent to determine which physically permissible outcome becomes actual, over and above the physical processes that constitute the agent and the realized history.

In discussing a specific choice between A and B, we will use decision for the point in the actual physical history at which A rather than B is established as the realized outcome. Prior intentions, reconsiderations, and apparent commitments are physical events too, but for this purpose they belong to the deliberative history leading to the outcome; they are not privileged as separate decisions. What counts as the outcome must, of course, be specified for the question under examination.

These definitions are analytical choices for the present discussion, not universal definitions used in philosophy or cognitive science. We will first review established conceptions of free will fairly, then explain the particular interpretation suggested by the physicalist Block Universe developed in this manuscript.

5.2 Physicalist free will

The first broad interpretation is that the conscious entity is entirely physical. Its deliberation is performed by physical processes, most plausibly processes occurring in the brain, and those processes obey the laws of physics.

Under this interpretation, there is no additional non-physical agent that has to intervene in the physical process in order for a choice to occur.

5.2.1 Physicalist-deterministic free will

In a deterministic physical theory, the complete physical state of the system together with the physical laws determines its subsequent physical states. A simplified representation is:

$$S(t0) + L \rightarrow S(t1)$$

For a conscious entity, those states include the physical processes associated with perception, memory, reasoning, deliberation, preferences, intentions, decisions, and actions.

There is no conflict between this picture and the Block Universe. Indeed, the Block Universe is naturally capable of representing such a deterministic history. The successive states of the entity form part of its world line:

$$S1 \rightarrow S2 \rightarrow S3 \rightarrow \cdots$$

The crucial point is that the Block Universe itself is indifferent to determinism. The Block Universe describes the complete four-dimensional history; it does not, by itself, specify whether the laws that generate or constrain that history are deterministic.

5.2.2 Physicalist-indeterministic free will

The same is true if the underlying physical laws contain an irreducibly random, or ontically random, component.

Instead of a physical state having only one possible successor, the laws might assign probabilities to several possible successors:

$$S(t0) + L \rightarrow \{S1, S1', S1'', \cdots\}$$

The Block Universe does not object to this. The actual four-dimensional universe simply contains the particular history that occurs.

This is consistent with the distinction developed in Section 4 between randomness and independence. A variable may be genuinely random and nevertheless have a definite value in the actual history.

The Block Universe specifies the actual history; it does not require the laws governing that history to be deterministic.

Thus, neither deterministic physicalism nor indeterministic physicalism is in conflict with the Block Universe merely by virtue of being deterministic or indeterministic.

5.3 Compatibilist free will

Compatibilism identifies free will with meaningful forms of agency exercised through a person's own reasons, values, preferences, and deliberation, rather than with a power to escape physical causation. On this account, an action may be free even when the processes producing it are entirely physical. Freedom may also depend on the absence of coercion or manipulation. Compatibilists therefore need not claim that an agent stands outside the laws of physics.

From the perspective adopted here, this is a defensible description of a genuine capability, but calling it free will risks confusing it with the stronger notion of an independent power to select an outcome. We prefer deliberative will for the physical capacity and perceived free will for the accompanying subjective experience. This is a terminological and interpretive preference, not a refutation of compatibilism.

The brain's organization, memories, preferences, conscious evaluation, and unconscious processing are all parts of one physical system. The detailed mechanism by which this system generates a subjective sense of agency is not fully understood. Nevertheless, it is reasonable within a physicalist account to treat that experience as emerging from the system, without claiming to have derived it from neuroscience.

The analogy with Section 2 is illuminating. Asymmetric physical records along an observer's world line help explain why time is experienced as flowing, even though the Block Universe requires no objectively advancing present. Likewise, the physical history contains representations of alternatives, deliberation, and the realized choice; an observer can experience this as exercising free will without that experience establishing an additional power outside the physical history.

$$\text{experienced flow of time} \leftrightarrow \text{asymmetric physical records}$$

$$\text{perceived free will} \leftrightarrow \text{physical records of deliberation and realized choice}$$

The analogy does not assert that the two experiences have identical neural mechanisms. Its point is ontological: a real subjective experience need not imply that the additional fundamental property attributed to it exists.

5.4 Libertarian free will

A stronger conception of free will holds that a person is genuinely capable of making alternative choices. In its strongest form, libertarian free will requires that the person be an ultimate source of the decision rather than merely the location at which a prior chain of physical causes terminates.

This raises the familiar question:

If a person's complete four-dimensional history contains deliberation followed by choice A, could that person have chosen B instead?

The answer depends on what is meant by could.

The Block Universe itself does not say that only one conceivable four-dimensional history exists. Consider two possible complete histories:

$$BA: \quad Spre \rightarrow A$$

$$BB: \quad Spre \rightarrow B$$

The two histories can be identical up to the decision event and differ only in the subsequent choice. Nothing about the Block Universe ontology itself prohibits such a pair of possible histories.

The Block Universe says that the actual universe is one complete four-dimensional history. It does not answer why the actual universe is BA rather than BB, nor does it by itself specify the physical laws or boundary conditions that determine which complete history is realized.

The Block Universe is not a theory of why this particular Block Universe exists rather than another possible Block Universe.

Consequently, the Block Universe should not be confused with determinism. Determinism is a claim about the dynamics of a physical theory. The Block Universe is an ontological description of spacetime.

$$\text{Block Universe} \neq \text{Deterministic Universe}$$

$$\text{Block Universe} \neq \text{Predetermined Universe}$$

A deterministic physical theory may imply that a particular prior state permits only one subsequent state. But that conclusion comes from the deterministic laws, not from the Block Universe concept itself.

Likewise, an indeterministic theory may permit multiple possible complete histories while the actual Block Universe contains only one of them.

5.5 Two meanings of “could have chosen differently”

The phrase “I could have chosen differently” therefore needs to be treated carefully. There are at least two distinct meanings.

5.5.1 Possible-history sense

There exists another physically possible complete history that is identical to the actual history up to the decision and in which the person chooses differently.

$$BA: \quad Spre \rightarrow A$$

$$BB: \quad Spre \rightarrow B$$

Under this interpretation, the Block Universe is entirely compatible with the statement that the person could have chosen differently.

5.5.2 Counterfactual-within-the-same-history sense

A stronger interpretation asks whether the person could have changed the actual outcome while leaving the actual four-dimensional history otherwise unchanged.

This is a substantially different proposition. The Block Universe does not require us to adopt it, and it is not necessary to establish that alternative complete histories are possible.

“Another possible history” is not the same thing as “another outcome within the same actual history.”

Once this distinction is made, the apparent contradiction between libertarian free will and the Block Universe becomes considerably less immediate.

5.6 Sourcehood and personhood

Another conception of free will focuses not on metaphysical indeterminacy but on sourcehood: whether the person is the genuine source of the action.

Every physical record in the universe has an antecedent history. The physical history of a human being can ultimately be traced backward through the history of life, the formation of the Earth, stellar and galactic evolution, and ultimately toward the earliest physical conditions of the universe.

$$\text{Universe} \rightarrow \text{galaxies} \rightarrow \text{stars} \rightarrow \text{planets} \rightarrow \text{life} \rightarrow \text{brains} \rightarrow \text{deliberation} \rightarrow \text{choice}$$

But the existence of this larger causal history does not eliminate the person as a distinct physical and causal system.

A person's decision can simultaneously be described at different levels. It can be a physical event caused by the state of the person's brain, while also being an action caused by the person's beliefs, memories, preferences, intentions, and deliberation.

These descriptions need not compete with one another. The person is a physical structure within the larger causal structure of the universe.

In this sense, personhood and sourcehood can be understood as properties embodied along the person's world line. The fact that the world line has physical antecedents extending back through the universe does not erase the causal reality of the person at the scale at which the person's decisions and actions occur.

The existence of antecedent causes extending throughout the history of the universe does not eliminate the person as a distinct physical and causal system within that history.

At this point the question begins to move from physics into philosophy. Physics can describe the physical processes constituting the person and the causal relations among them. Whether that constitutes the kind of "ultimate authorship" required by a particular philosophical conception of free will is a further philosophical question.

5.7 Non-physical and dualist conceptions

A fundamentally different proposal is that conscious agency is not entirely physical. A soul, mind, or other non-physical agent could be proposed as an additional source of choice.

The Block Universe does not itself disprove such a proposal. The issue is different: without a theory describing the proposed non-physical entity and its interaction with the physical universe, the proposal does not provide a physical explanation of the choice.

For example, suppose we write:

$$\text{brain state} \rightarrow \text{non-physical agent chooses} \rightarrow \text{brain state changes}$$

We then need to ask what the state of the non-physical agent is, what rules govern its behavior, how it interacts with physical matter, and what observable consequences distinguish this account from a purely physical explanation.

Simply asserting that "the soul made the choice" relocates the unexplained mechanism rather than specifying it.

A non-physical theory could certainly be developed in principle, but it would require additional structure before it could function as an explanatory physical theory. Otherwise, sufficiently unconstrained metaphysical agency can be invoked to explain almost any outcome, making it difficult to distinguish the hypothesis empirically.

This limitation is not imposed by the Block Universe. It is a general requirement for constructing a testable physical explanation.

5.8 Free will and the independence of other conscious entities

Our original definition included another interesting condition: that an individual's deliberation should be independent of the internal deliberations of other conscious entities.

This should be distinguished from statistical correlation.

As established in Section 4, the existence of a correlation does not by itself imply a causal interaction. In particular:

$$\text{correlation across spacetime} \neq \text{interaction across spacetime}$$

The same distinction applies to human decisions. If two people's decisions are correlated, that does not by itself establish that one person's deliberation has overridden or controlled the other's deliberation.

For free will, the more relevant question is whether another agent has bypassed or coerced the person's own deliberative process.

Thus statistical dependence should not automatically be identified with loss of agency. This is particularly important in the present framework, because the Block Universe permits correlations among events at different locations and times without requiring a corresponding physical signal propagating between them.

5.9 The Block Universe does not determine the origin of the actual history

We can now state an important limitation of the framework.

The Block Universe tells us that the actual universe is a complete four-dimensional history. It does not tell us why that particular history exists rather than another possible history.

Suppose two complete histories are physically possible:

$$BA = \{\text{all events, with choice A}\}$$

$$BB = \{\text{all events, with choice B}\}$$

The Block Universe framework does not contain an additional principle that says: “BA must exist rather than BB.”

That question belongs to whatever deeper theory specifies the physical laws, boundary conditions, probability structure, or other principles governing the universe.

This is analogous to the role of the Block Universe in our discussion of quantum mechanics. In Section 4, the Block Universe provided a framework in which complete four-dimensional histories could contain correlations without requiring instantaneous causal influence. It did not itself derive the particular quantum probability distribution

$$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(\mathbf{x} \cdot \mathbf{y})] / 4.$$

Likewise, in the present discussion, the Block Universe provides the four-dimensional ontology but does not itself determine why a particular person's world line contains one choice rather than another.

5.10 A useful taxonomy of free will

Conception	What makes an action free?	Compatibility with the Block Universe
Physicalist-deterministic	Physical deliberation produces the action under deterministic laws.	Compatible. The Block Universe is indifferent to determinism.
Physicalist-indeterministic	Physical deliberation produces the action while the underlying laws contain genuine indeterminacy.	Compatible. The actual Block Universe contains one history permitted by the laws.
Compatibilist	The action flows from the person's own reasons, memories, preferences, intentions, and deliberation.	Compatible. The deliberative process is itself part of the person's world line.
Libertarian	The person is genuinely capable of alternative choices and is an appropriate ultimate source of the decision.	Not prohibited by the Block Universe. Alternative complete histories can be conceived without requiring the Block Universe to explain why the actual one is realized.
Sourcehood	The action originates through the person's own deliberative and motivational structure.	Compatible. A person remains a distinct causal system within the larger four-dimensional history.
Dualist / non-physical	A non-physical component participates in or controls the decision.	Not ruled out by the Block Universe, but requires an additional theory describing the non-physical agent and its interaction with the physical universe.

5.11 What the Block Universe actually implies about free will

Under the definition used throughout this manuscript, the actual Block Universe is the complete physical history of reality: one particular realization consistent with the laws of physics. It contains the observer, the brain, the deliberation, any genuinely random events, and the realized decision. It is not a stage on which an additional agent stands outside the history and selects what will happen.

This makes the phrase independent control problematic unless its meaning is specified. Independent of what? If it means independent of the complete physical history or able to suspend physical laws, it is incompatible with the physicalist Block Universe being considered here. If it refers instead to a physically lawful process inside the history, that process is already part of the universe; it may be described as agency or regulation, but it does not introduce a separate freedom to choose which complete history is actual.

Neither determinism nor indeterminism changes this reasoning. Deterministic laws may permit one outcome given a complete prior state; indeterministic laws may permit several with specified

probabilities. Randomness does not confer control over the random result upon the agent. In either case the actual Block Universe contains the outcome that occurred.

This does not make deliberation unreal or causally irrelevant. The agent's reasoning is part of the physical account of the realized action. Nor does it imply that the action was externally coerced: a person and the physical processes constituting that person are not two competing sources of causation.

The observer is part of the realized history, not an additional selector of that history. Deliberative will is physically real; the stronger freedom commonly attributed to it is not established by the experience of choosing.

Here, to say that the laws of physics govern the decision means that the complete history must satisfy physical constraints, not that the laws themselves are conscious agents that actively choose outcomes. The Block Universe does not supply an explanation of why this permissible history exists rather than another.

5.12 The central conclusion

The distinction between deliberative will and fundamental free will allows us to state the present interpretation more precisely. An observer may consciously consider A and B, revise intentions, and ultimately realize A. Every part of that process is contained in the physical history. The decision, in the sense used here, is the realized A rather than B; earlier tentative commitments are simply events within the deliberation leading to it.

The observer may experience the process as an independent act of choosing. From the observer's immediate perspective, that experience may be indistinguishable from what the observer calls genuine free will. But the experience alone does not establish an additional power to select or revise the complete physical history.

This is parallel to our account of temporal flow. Physical records underpin an observer's experience of time passing without requiring an objectively moving present. Similarly, the physical processes and records of deliberation and choice can underpin an experience of free will without requiring a separate selector of physical reality.

Our preferred interpretation is that deliberative will is a real physical capability, while perceived free will is a real subjective experience. Neither entails fundamental freedom from the physical history or its governing laws.

The conclusion does not depend on determinism. It also does not purport to derive the subjective experience of agency from a completed theory of consciousness. It is an ontological interpretation grounded in the physicalist Block Universe assumed here, not a theorem of relativity alone or a demonstration that every philosophical account of agency is incoherent.

Alternative philosophical definitions, including compatibilist and sourcehood accounts, can remain meaningful descriptions of internal physical agency. What this interpretation rejects is the inference from those descriptions to an extra power of selection outside the complete physical history.

5.13 Transition to the next question

This leaves us with a more precise problem than the one with which we began.

How do physical processes within an observer produce the experience of choosing, and what can that experience legitimately tell us about reality?

That question takes us beyond the ontology of a complete four-dimensional history and toward the physical structure of observers, information, consciousness, and the relationship between deliberation and realized outcomes.

The same conceptual strategy that proved useful in the quantum-mechanical discussion can be applied here: distinguish the existence of a correlation or a complete history from the existence of an external causal influence.

global consistency ≠ external coercion

future existence ≠ absence of agency

6 The Block Universe and Time Travel

Backward travel, causal interactions with the past, logical self-consistency, and closed timelike curves

The central issue raised by backward time travel is not whether a traveler's worldline closes, but whether all events—including interactions with an earlier history—can belong to one logically consistent Block Universe.

6.1 Time Travel as a Question of Causal Structure

In special and general relativity, a massive observer follows a future-directed timelike worldline. Proper time increases along that worldline. Relativistic time dilation allows two observers to accumulate different amounts of proper time between meetings; this familiar effect is sometimes called travel into the future. The more challenging question is whether an observer can reach the chronological past of an event from which they departed.

Let A be the traveler's departure event and B the arrival event. The chronological past $I^-(A)$ consists of events from which a future-directed timelike path to A exists. In flat spacetime this is the interior of A's past light cone. Thus $B \in I^-(A)$ means that B can causally precede A by a timelike route. The causal past $J^-(A)$ also includes events connected to A by lightlike paths.

An arrival at a numerically smaller coordinate time is not enough to establish physical travel into the past. Coordinate labels can be changed. The invariant question is whether B lies in the causal or chronological past of A while the traveler reaches B by following a locally future-directed causal path from A. General relativity admits mathematical spacetimes with such nontrivial global causal relationships, although their physical realization remains uncertain.

Crucially, the traveler's own worldline need not close. The observer can depart at A, arrive at B, and continue onward without ever returning to the departure event. Backward time travel and a closed traveler worldline are therefore different concepts.

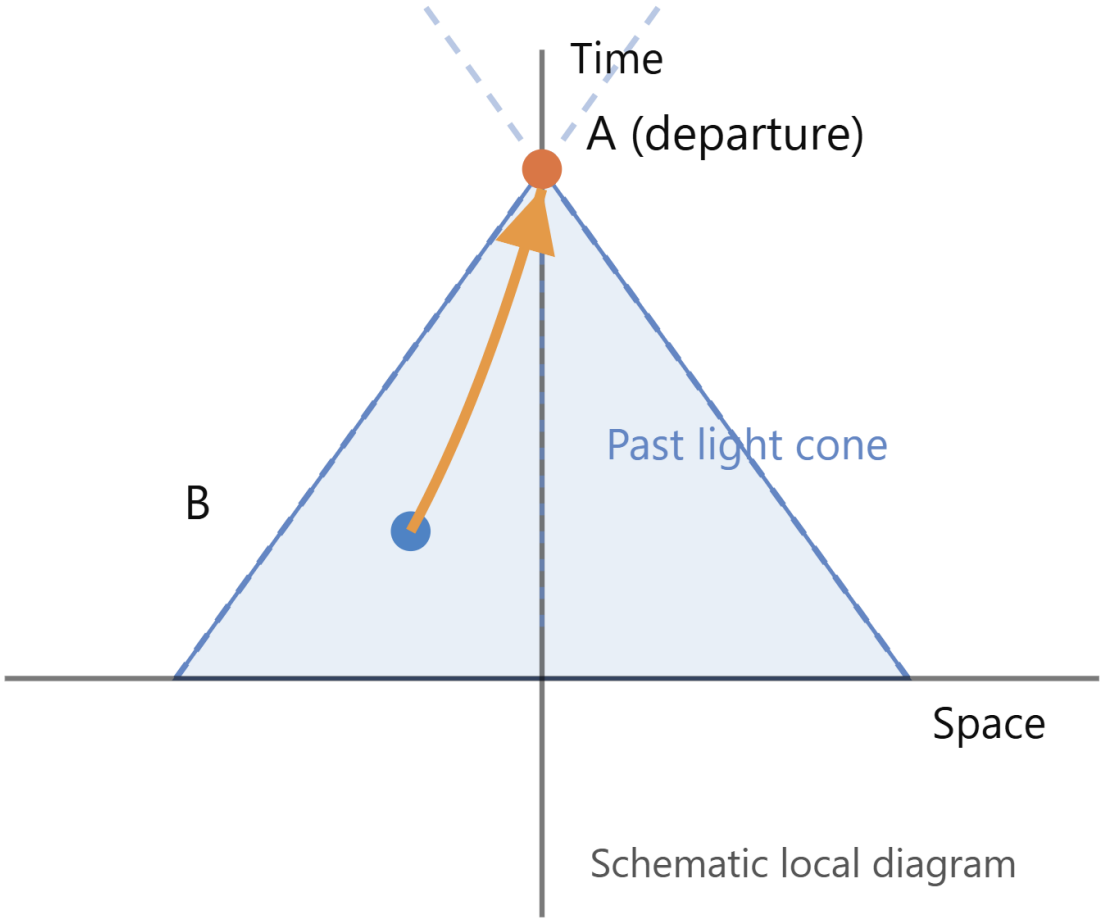
6.2 Open Worldlines and Encounters with the Earlier History

After arriving at B in the chronological past of A, the traveler can follow an open worldline. Several possibilities arise. The traveler might avoid any interaction with their younger self. They might send a message to their younger self. Or they might interact with other people and objects whose subsequent histories affect the circumstances of departure at A.

A meeting between older and younger versions of the same person does not require the older traveler to revisit the departure event A. The two versions can be present in the same region of spacetime at an event distinct from A, or communicate across distinct events. If they literally occupy the same event, the traveler's full worldline has a self-intersection there; it need not be a globally periodic worldline. A message exchanged across separate events can leave the material worldline non-self-intersecting.

The relevant condition is that the earlier person's memories and records, the later departure, and the older traveler's actions all agree as parts of one physical history. If the younger observer remembers receiving a warning, then the older observer who eventually sends it cannot belong to an otherwise identical history in which the warning was never received.

Even when there is no direct encounter, indirect causal effects may matter. Avoiding one's younger self removes an obvious opportunity for contradiction, but it does not independently prove the complete history consistent. Conversely, interaction with one's younger self is not itself paradoxical.



6.3 The Grandfather Paradox Without a Closed Traveler Worldline

The grandfather paradox is an example of backward time travel with a potentially open traveler worldline. A traveler departs at A, arrives at B in an earlier era, and attempts to kill an ancestor before the ancestor has the descendants required for the traveler's own birth. The traveler need not return to A, and the observer's material trajectory need not form a closed loop.

The contradiction arises from the stipulated facts: the traveler exists and makes the journey, yet the intervention is specified to prevent the conditions necessary for that same traveler to exist. Both claims cannot be true of one complete history under the assumed genealogy and circumstances.

The paradox is often narrated as though a previously completed past were subsequently modified. In the Block Universe developed in Section 1, however, departure, arrival, intervention, and their consequences are all events in one four-dimensional history. There is no second temporal process in which an already existing past is rewritten.

This shows why the grandfather paradox is more general than a paradox involving a traveler who returns to the same spacetime event. Its essential feature is a contradictory causal specification, not geometric closure of the observer's worldline.

6.4 Logical Self-Consistency as a Condition for Existence

The preceding examples motivate a principle broader than time travel: logical self-consistency is a necessary condition for physical existence. A complete history containing mutually contradictory facts cannot be an actual physical history.

$$\text{Exists}(H) \Rightarrow \text{LogicallyConsistent}(H).$$

The implication is one-way. Logical consistency does not establish that a hypothetical history exists, nor that it satisfies Einstein's field equations, quantum theory, conservation laws, or appropriate boundary conditions. A consistent imagined history may still be physically impossible.

Applied to the grandfather paradox, this principle does not posit a special force that jams the weapon or arranges a last-minute rescue. It states only that the contradictory set of events cannot constitute one actual history. A specific consistent history may contain a failed attempt, different ancestry, or some other outcome, but the principle by itself does not predict which outcome occurs.

The same requirement applies whether the traveler's worldline is open or closed, and whether causal contact with the younger self is direct, indirect, or absent. Time travel makes the constraint conspicuous; it does not create the requirement.

6.5 Permitted Causal Paths Versus Realized Physical Histories

General relativity describes a spacetime metric and the causal paths it permits. The actual Block Universe additionally contains the matter, radiation, observers, interactions, and physical records that are realized. A geometrically permitted trajectory that no object follows is not an extra material history or a second copy of an observer.

If the traveler goes from A to B along a future-directed timelike route while B lies in $I^-(A)$, there must also be a future-directed timelike route from B to A. Concatenating these routes establishes that the geometry admits a closed timelike path. This conclusion does not require the traveler to take the return route.

Thus an open actual traveler worldline can coexist with closed timelike curves admitted by the spacetime geometry. The existence of a geometric curve is not the same claim as the existence of a physical observer traversing it. The distinction matters when interpreting general relativity, but the ontological analysis of paradoxes concerns the actual events and their consistency.

Where an actual traveler affects their own earlier history, the relevant causal relationships can constrain that history without requiring the traveler's own worldline to close.

6.6 Closed Timelike Curves as a Special Geometrical Case

A closed timelike curve (CTC) is a future-directed timelike curve that returns to the same spacetime event. Writing the curve as $\gamma(\tau)$, with τ proper time, its geometric closure can be represented by

$$\gamma(\tau + T) = \gamma(\tau),\ T > 0.$$

Proper time increases locally throughout the circuit. The curve is globally closed even though the observer never reverses their own local temporal direction or exceeds the speed of light.

Gödel's rotating-universe solution provides a classic mathematical example. Certain idealized Kerr extensions and hypothetical time-shifted wormhole constructions also motivate discussions of chronology violation. These geometries are not evidence that realistic time machines exist.

The conceptual example A → B → A can be read in two ways. If the traveler actually takes both future-directed segments, the full path is closed. If the traveler takes only A → B and proceeds elsewhere, their path remains open, although a geometrically permitted B → A route may still exist. In neither case should a merely permitted return route be mistaken for an additional physical traveler.

6.7 A Closed Worldline and the Traveler's Physical State

Geometric closure identifies the same spacetime event after a circuit, but a physical traveler also has internal degrees of freedom: a clock reading, memories, biological state, and interactions with the environment. These cannot be assigned arbitrarily to different traversals of the same event.

For a single-valued physical state S transported around a genuinely periodic worldline, consistency requires

$$S(\tau + T) = S(\tau).$$

Equivalently, if F_T represents the state evolution around a circuit, a consistent periodic state is a fixed point: $F_T(S) = S$. A given physical evolution law need not possess such a state. Geometry alone does not guarantee a realizable material traveler.

This explains why an ordinary-looking B → A journey can seem to create two copies at A. In the idealized smooth periodic case, the returning and departing portions are the same locally identified history, not two independently specified bodies that must occupy the same event with conflicting states. A self-intersecting but nonperiodic trajectory, or a model with multiple streams meeting, requires a more detailed matter description; geometric coincidence alone does not identify all internal states.

The state-matching condition is therefore a strong constraint on a realized closed physical trajectory. It is not a requirement that every open-worldline traveler returning to an earlier era become a periodic physical system.

6.8 Proper Time, Memory, and the Experienced Direction of Time

Section 2 explained the experienced direction of time in terms of asymmetric physical records. An observer remembers events represented in earlier physical states but does not ordinarily possess corresponding records of later states. This information asymmetry is linked to thermodynamic conditions, rather than to an external movement of a universal present.

An observer on an open backward-time journey continues to experience increasing proper time. The same local statement applies along a CTC. In either case, a journey to the chronological past of departure need not feel like time running backward.

Whether an observer could detect the global closure of their own worldline is a separate question. Some spacetime configurations may offer observable evidence of unusual causal structure; others may not. The mere subjective experience of a succession of remembered states does not establish that a worldline is globally open.

One must not assume that memory information strictly increases at every point along every worldline. Forgetting, erasure, equilibration, and exchange with the environment are physical possibilities. Nonetheless, the requirement of a globally consistent state around a realized periodic curve can be highly restrictive for an ordinary memory-forming organism.

6.9 Thermodynamics and Recurrent Physical States

Ordinary biological activity and memory formation typically involve dissipative processes. If a physically isolated system is required to return to precisely the same state after one circuit, then a strictly increasing single-valued entropy around that circuit is incompatible with closure.

$$S_{entropy}(\tau + T) = S_{entropy}(\tau),$$

whereas

$$dS_{entropy}/d\tau > 0$$

everywhere would imply a contradiction.

This is a conditional incompatibility, not a general theorem excluding every CTC. Thermodynamic entropy increase is statistical; local decreases and transfers of entropy between subsystems and environments must be treated within the full physical model. Nor is there necessarily a globally defined time coordinate or universal entropy function that increases on every causal trajectory in a chronology-violating spacetime.

A closed parameterized curve may be traversed repeatedly in its mathematical description: $\gamma(\tau + nT) = \gamma(\tau)$. These repeated parameter values do not, by themselves, create infinitely many new spacetime events, independent copies of the traveler, or repeated deposits of heat. Distinct matter streams or physical interactions, if present in a proposed model, must still be accounted for.

The appropriate conclusion is modest: thermodynamics places additional constraints on physically realizable closed trajectories, particularly those involving ordinary observers, but it does not by itself convert every proposed time-travel history into a logical contradiction.

6.10 Global Consistency Without a Dynamical Intervention

Section 3 described physical laws as global relationships satisfied by a complete four-dimensional history, rather than necessarily as step-by-step instructions executed by an external clock. This viewpoint makes the time-travel puzzle less mysterious.

If the actual Block Universe contains an older traveler sending a message to a younger self, the reception, the later memories, the departure, and the transmission are all parts of one history. Nothing is added to a previously finished past. A contradictory description is not a real history awaiting repair.

This resembles the interpretation proposed in Section 4 for globally correlated quantum measurement settings and outcomes. In that discussion, global correlations need not be generated by superluminal signals or by a mechanism anticipating future choices. Here, consistent relationships between earlier and later events similarly need not be produced by a universe that detects paradoxes and intervenes to prevent them.

The analogy is conceptual, not a derivation. The Block Universe does not calculate which consistent time-travel history occurs, establish the dynamics of a time machine, or derive the statistical predictions of quantum mechanics. It offers an ontological interpretation of histories that are otherwise physically admissible.

6.11 Physical Possibility and the Central Conclusion

General relativity admits mathematical spacetimes with nontrivial causal structure, including closed timelike curves. Whether realistic matter and quantum effects permit the formation or traversal of such regions remains unresolved. The logical self-consistency principle neither proves time travel possible nor substitutes for the dynamical laws that any proposed geometry and matter configuration must obey.

The grandfather paradox illustrates the central point without requiring the traveler's own worldline to close. Backward time travel may create causal relationships with an observer's earlier history, and those relationships must form one consistent set of physical facts. The traveler may interact with a younger self, influence earlier events indirectly, or avoid such interactions; none of these alternatives escapes the general requirement of logical consistency.

Closed timelike curves are a special geometric manifestation of chronology violation. A spacetime can admit them without any observer actually traversing one, while an observer who does follow a genuinely closed physical worldline faces additional state-matching and thermodynamic constraints.

The Block Universe does not portray time travel as rewriting the past. It portrays any physically realized journey, its causes, its interactions, and its consequences as parts of a single four-dimensional history. Logical consistency

is necessary for that history to exist; physical laws determine which consistent histories, if any, are realizable.

Appendix A. CHSH and the Explicit Measurement-Dependent Local Hidden-Variable Model

This appendix contains the technical development underlying the discussion in Section 4. It is intentionally separated from the main narrative so that the conceptual argument can proceed without requiring the reader to follow the full CHSH algebra. The appendix derives the CHSH bound for local hidden-variable models, identifies the precise role of measurement independence, and presents the explicit measurement-dependent local hidden-variable model used in the main text.

The appendix supplies the mathematical support for the conclusions summarized in Section 4. It does not by itself establish that the Block Universe is the physical mechanism responsible for measurement dependence; that remains an open physical question discussed in the main text.

A.1 The CHSH Experiment

The CHSH formulation makes the logical structure of Bell's theorem especially transparent. Instead of considering a single measurement setting at each location, Alice has a choice between two settings, a and a', while Bob has a choice between two settings, b and b'. Each measurement produces an outcome of either +1 or −1.

For any pair of settings, define the correlation

$$E(a,b) = \langle AB \rangle.$$

The CHSH combination is then

$$S = E(a,b) + E(a,b') + E(a',b) - E(a',b').$$

The remarkable feature of CHSH is that a local hidden-variable model satisfying measurement independence places a strict upper bound on this quantity.

A.2 The Local Deterministic CHSH Bound

Begin with the deterministic special case:

$$A = A(a,\lambda), \quad B = B(b,\lambda),$$

where every outcome is either +1 or −1.

For a particular value of λ, define

$$S(\lambda) = A(a,\lambda)B(b,\lambda) + A(a,\lambda)B(b',\lambda) + A(a',\lambda)B(b,\lambda) - A(a',\lambda)B(b',\lambda).$$

Factor this expression:

$$S(\lambda) = A(a,\lambda)[B(b,\lambda)+B(b',\lambda)] + A(a',\lambda)[B(b,\lambda)-B(b',\lambda)].$$

Because $B(b,\lambda)$ and $B(b',\lambda)$ can each only be $+1$ or -1, one of the two bracketed terms must be zero and the other must be either $+2$ or -2. Therefore

$$|S(\lambda)| = 2.$$

This is an important point: the local algebra itself does not yet use measurement independence. The bound arises for each individual hidden-variable state.

A.3 Where Measurement Independence Enters CHSH

Under measurement independence, the same probability distribution $\rho(\lambda)$ is used for all four setting combinations:

$$E(a,b) = \int d\lambda\, \rho(\lambda)\, A(a,\lambda)B(b,\lambda),$$

$$E(a,b') = \int d\lambda\, \rho(\lambda)\, A(a,\lambda)B(b',\lambda),$$

$$E(a',b) = \int d\lambda\, \rho(\lambda)\, A(a',\lambda)B(b,\lambda),$$

$$E(a',b') = \int d\lambda\, \rho(\lambda)\, A(a',\lambda)B(b',\lambda).$$

Because the same distribution appears in all four expressions, they can be combined:

$$S = \int d\lambda\, \rho(\lambda)\, S(\lambda).$$

Since $|S(\lambda)| = 2$ for every λ, it follows that

$$|S| \leq 2.$$

This is the CHSH inequality.

The exact point at which measurement independence enters is therefore extremely specific. It is **not** the algebraic identity that gives $|S(\lambda)|=2$. It is the assumption that the four experimentally different setting combinations can all be represented as averages over the **same underlying distribution** $\rho(\lambda)$.

If instead the distribution depends on the settings, we have

$$\rho(\lambda|a,b), \quad \rho(\lambda|a,b'), \quad \rho(\lambda|a',b), \quad \rho(\lambda|a',b'),$$

and the four terms cannot in general be combined into a single integral of $S(\lambda)$. The step producing $|S|\leq 2$ is then unavailable.

$$\text{Measurement independence} \rightarrow \text{one common } \rho(\lambda) \rightarrow \text{CHSH bound.}$$

This is the mathematical opening on which the remainder of the proposal rests.

A.4 The Quantum Prediction

For a pair of spin-½ particles prepared in the singlet state, quantum mechanics predicts the correlation

$$E_{QM}(a,b) = -\cos(\theta_{ab}),$$

where θ_{ab} is the angle between the measurement directions.

Choose, for example,

$$a = 0°, \quad a' = 90°, \quad b = 45°, \quad b' = -45°.$$

The corresponding quantum correlations give

$$|S_{QM}| = 2\sqrt{2}.$$

Since

$$2\sqrt{2} > 2,$$

the quantum prediction violates the CHSH bound.

The experimental significance is straightforward: measurements reproduce correlations incompatible with the assumptions used to derive the local, measurement-independent CHSH inequality.

A.5 Deterministic Notation Does Not Require Deterministic Ontology

At this stage it is important not to overinterpret the notation $A(a,\lambda)$ and $B(b,\lambda)$. The deterministic treatment is mathematically convenient because it makes the CHSH logic transparent. The broader proposal does not require that nature be deterministic at the hidden-variable level.

A more general local stochastic model would instead specify conditional probabilities such as

$$P(A|a,\lambda), \quad P(B|b,\lambda).$$

Locality can then be represented by the factorization

$$P(A,B|a,b,\lambda) = P(A|a,\lambda)P(B|b,\lambda).$$

Measurement dependence concerns the relationship between λ and the settings, not whether the response functions themselves are deterministic. The proposal therefore remains compatible with genuinely stochastic underlying physics.

A.6 The Constructive Question

The CHSH analysis has established the logical opening, but a logical opening is not yet a physical model. To make the proposal more substantial, we need to show that a measurement-dependent local hidden-variable construction can actually reproduce the quantum correlations.

Such constructions are known in the foundations of physics literature. A particularly simple example, associated with Brans and discussed explicitly by Michael Hall, uses a hidden variable with two binary components. The model is useful here not because it constitutes the complete physical theory being proposed, but because it demonstrates that relaxing measurement independence is mathematically sufficient to construct local response functions reproducing the singlet statistics.

A.7 The Hidden Variable

Let the hidden variable be

$$\lambda = (\lambda_1, \lambda_2),$$

where

$$\lambda_1, \lambda_2 \in \{+1, -1\}.$$

There are therefore four possible values of λ.

Define the local measurement responses simply as

$$A(x,\lambda) = \lambda_1, \quad B(y,\lambda) = \lambda_2.$$

These response functions are manifestly local in the relevant sense: Alice's result depends on her local hidden-variable component and not on Bob's measurement direction, while Bob's result depends on his local hidden-variable component and not on Alice's measurement direction.

A.8 The Setting-Dependent Hidden-Variable Distribution

The essential ingredient is the distribution assigned to the hidden variable. Instead of assuming a setting-independent distribution, define

$$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(x\cdot y)] / 4.$$

Here x and y are the two measurement directions, represented as unit vectors.

This equation contains the entire departure from measurement independence. The local response functions remain simple. All of the angular dependence required to reproduce the quantum correlation is placed in the conditional distribution of the hidden variable.

A.9 Validity of the Probability Distribution

A probability distribution must be non-negative and must sum to one.

Because $\lambda_1\lambda_2=\pm 1$ and $-1\le x\cdot y\le 1$, the numerator satisfies

$$0 \le 1 - \lambda_1\lambda_2(x\cdot y) \le 2.$$

Thus

$$\rho(\lambda|x,y) \ge 0.$$

Summing over the four possible values of λ gives

$$\Sigma\lambda\ \rho(\lambda|x,y) = [4 - (x\cdot y)\Sigma_\lambda\lambda_1\lambda_2]/4 = 1,$$

because the four values contain equal numbers with $\lambda_1\lambda_2\}=+1$ and -1.

The construction is therefore a valid conditional probability distribution.

A.10 The Local Marginals

A crucial property of the model is that Alice's marginal statistics remain independent of Bob's setting, and vice versa.

Since Alice's outcome is $A=\lambda 1$, sum over $\lambda 2$:

$$P(A=+1|x,y) = 1/2, \quad P(A=-1|x,y) = 1/2.$$

Likewise,

$$P(B=+1|x,y) = 1/2, \quad P(B=-1|x,y) = 1/2.$$

Thus, neither observer can infer the distant measurement setting merely by examining the local outcome statistics.

This is important conceptually. The model is measurement-dependent, but the dependence does not provide an operational faster-than-light signaling channel.

A.11 Reproducing the Singlet Correlation

Because the outcomes are

$$A = \lambda_1, \quad B = \lambda_2,$$

their product is

$$AB = \lambda_1\lambda_2.$$

The predicted correlation is therefore

$$E(AB|x,y) = \Sigma\lambda\ \rho(\lambda|x,y)\ \lambda_1\lambda_2.$$

Substituting the conditional distribution gives

$$E(AB|x,y) = \Sigma\lambda\ [1 - \lambda_1\lambda_2(x\cdot y)]\ \lambda_1\lambda_2/4.$$

The first term sums to zero because the four hidden-variable states contain equal numbers of positive and negative products. The second term gives

$$E(AB|x,y) = -x\cdot y.$$

For unit measurement vectors this is exactly

$$E(AB|x,y) = -\cos(\theta_{xy}),$$

which is the singlet correlation predicted by quantum mechanics.

A.12 Reproducing the Full Singlet Joint Probabilities

The construction is stronger than reproducing only the average correlation. Since $A=\lambda_1$ and $B=\lambda_2$, the joint probability of outcomes $a,b\in\{\pm 1\}$ is

$$P(A=a,B=b|x,y) = [1 - ab(x\cdot y)]/4.$$

This is precisely the singlet-state joint probability distribution for spin measurements along directions x and y.

Thus, the simple model does not merely produce the correct value of $E(AB)$. It reproduces the complete two-outcome probability distribution for the singlet measurement.

A.13 What the Explicit Model Demonstrates

The model demonstrates something very specific and important:

Local response functions + measurement dependence → exact singlet statistics.

It therefore establishes that Bell-violating correlations do not logically require the response functions themselves to contain a dependence on the distant measurement setting.

However, the model should not be described simply as an ordinary Bell-local hidden-variable theory. Bell locality, in the full sense relevant to the theorem, includes the statistical independence assumption. Because the distribution $\rho(\lambda|x,y)$ depends on the settings, the model violates measurement independence.

The most precise description is therefore a **measurement-dependent local hidden-variable model**, or a model with **local response functions** and setting-dependent hidden-variable statistics.

A.14 The Model Is Not Yet the Block Universe

The mathematical construction should not be confused with the physical interpretation proposed in this manuscript.

The equations above demonstrate that a particular conditional probability distribution can reproduce the singlet correlations while the response functions remain local. They do not, by themselves, explain *why* that conditional distribution has the required form.

That is where the Block Universe enters conceptually. The proposal is that the setting-dependent distribution may be understood not as a mysterious dynamical influence from a future measurement into the past, but as a property of the complete four-dimensional physical history.

The mathematical model shows that such correlations are possible. The Block Universe supplies the spacetime picture within which the existence of such correlations can be considered without automatically interpreting them as causal signals traveling backward through time.

A.15 Correlation Across Spacetime Is Not Interaction Across Spacetime

Suppose a complete physical history contains the source preparation, the hidden variable λ, Alice's eventual setting x, Bob's eventual setting y, and the corresponding outcomes.

$$H = \{\text{preparation}, \lambda, x, y, A, B, \text{measurement records}, \text{spacetime geometry}, \ldots\}$$

In a Block Universe description, these are not pieces of reality that must be created one after another. They are components of one four-dimensional history.

Consequently, a statistical relationship between λ and later settings does not automatically imply that the later settings physically caused λ. It may instead be a constraint on which complete histories occur with which probabilities.

$$\text{Statistical dependence} \neq \text{dynamical causation.}$$

This distinction is central to the proposal. Without it, the rejection of measurement independence immediately appears to require retrocausal signaling. With it, the possibility of a globally correlated four-dimensional history becomes conceptually distinct from a signal propagating backward through time.

A.16 Does This Mean the Future Determines the Past?

That language is tempting but potentially misleading.

In the ordinary dynamical picture, one asks whether a later event causes an earlier event. The Block Universe picture does not require that question to have the same role. Instead, it asks whether the complete spacetime solution satisfies a set of global consistency conditions.

Consider an ordinary four-dimensional geometric object. Its distant parts can stand in exact mathematical relationships to one another without one part exerting a force on another. The relationship is a property of the whole object.

The proposal applies an analogous idea to physical histories. The correlation between a hidden variable and measurement settings is not necessarily a new causal process. It can instead be part of the structure selecting which complete physical history is realized.

This does not prove that nature actually works this way. It identifies the conceptual possibility that must be examined by a complete theory.

A.17 Ontic Randomness Does Not Require Statistical Independence

Another important distinction concerns randomness.

If λ is genuinely random, it does not follow that λ must be statistically independent of every other variable in the universe. Randomness and independence are different mathematical concepts.

A variable can have a nontrivial probability distribution while remaining correlated with another variable. In the present construction, the hidden variable is not assigned a single fixed value in advance from the perspective of the observer. Instead, different complete histories occur with different probabilities, and those probabilities depend on the measurement settings that form part of the complete history.

Thus, the proposal does not have to say that quantum randomness is merely ignorance about a pre-existing deterministic state. It can retain genuine probabilistic structure while rejecting the stronger assumption that the random variable must be statistically independent of later measurement choices.

A.18 The Entire Angular Dependence Resides in the Hidden-Variable Distribution

The explicit model makes the structure particularly transparent.

The response functions are simply

$$A(x,\lambda)=\lambda_1, \quad B(y,\lambda)=\lambda_2.$$

They contain no dependence on the distant measurement direction.

All of the angular dependence required to reproduce the quantum correlation appears in

$$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(x\cdot y)]/4.$$

The quantum angular correlation is therefore represented as a property of the distribution of complete histories rather than as a nonlocal modification of a local response function.

A.19 No-Signaling Is Preserved

The model's measurement dependence should not be confused with operational signaling.

Although the hidden-variable distribution depends jointly on *x* and *y*, the local marginal distributions remain

$$P(A=\pm1|x,y)=1/2,$$

$$P(B=\pm1|x,y)=1/2.$$

Therefore, Alice's local statistics do not reveal Bob's setting, and Bob's local statistics do not reveal Alice's setting.

This distinction is important for the proposed interpretation. The aim is not to introduce a hidden communication channel that transmits usable information faster than light. The aim is to account for the observed joint correlations without requiring such a channel.

A.20 The Price of the Proposal: Measurement Independence Is a Real Assumption

The measurement-dependent construction does not make Bell's theorem disappear. On the contrary, it clarifies exactly what Bell's theorem tells us.

If measurement independence is imposed, the CHSH inequality follows for the local hidden-variable framework described above. Quantum mechanics violates that inequality. Therefore, a local hidden-variable account reproducing the quantum correlations must relax at least one of the assumptions entering the Bell derivation.

The present proposal chooses to relax measurement independence.

That is a substantive physical commitment, not a mathematical trick. The question is whether the resulting commitment can be given a natural physical interpretation and developed into a complete theory without sacrificing the empirical successes of quantum mechanics.

A.21 The Central Proposal, Sharpened

The proposal developed in this section can now be stated more precisely.

A local hidden-variable model may reproduce the exact singlet correlations if the hidden-variable distribution is allowed to depend statistically on the measurement settings:

$$\rho(\lambda|x,y) \neq \rho(\lambda).$$

The local response functions may remain of the form

$$A=A(x,\lambda), \quad B=B(y,\lambda),$$

with no superluminal dependence of one local outcome on the distant setting.

The Block Universe interpretation proposed here is that the setting dependence may represent a correlation among components of one complete four-dimensional physical history rather than a dynamical influence propagating from the future into the past.

A.22 The Observer Still Experiences a Sequence

None of this changes the observer's ordinary experience of time.

Along Alice's worldline, there is still a sequence of physical states:

$$S1 \rightarrow S2 \rightarrow S3.$$

At one state, she prepares the experiment. At a later state, she encounters a particular measurement setting and records an outcome. At still later states, she remembers that outcome and incorporates it into subsequent reasoning.

The fact that the complete history contains the eventual outcome does not mean that Alice experiences that outcome before the measurement event. Her physical state at the earlier point does not ordinarily contain a usable record of the later result.

This preserves the distinction developed earlier between the four-dimensional existence of events and the sequential information structure of an embedded observer.

A.23 What the Block Universe Does and Does Not Explain

At this point the role of the Block Universe must be kept carefully bounded.

It provides a conceptual framework in which:

- past and future events can be components of one four-dimensional history;
- statistical correlations need not automatically be interpreted as causal signals;
- measurement events can be treated as local physical events within the spacetime structure;
- the observer's sequential experience remains compatible with a complete spacetime description.

But the Block Universe does **not**, by itself, derive the specific probability distribution

$$\rho(\lambda|x,y) = [1 - \lambda_1\lambda_2(x\cdot y)]/4.$$

Nor does it yet provide a dynamical or variational principle from which that distribution follows. That is one of the central unresolved problems for the proposed framework.

A.24 The Joint Distribution as the More Fundamental Object

The conventional measurement-independent picture begins with an underlying distribution

$$\rho(\lambda)$$

and then treats the measurement settings as independently selected variables.

The present proposal instead suggests that the more fundamental object may be a joint distribution over the complete set of relevant physical variables:

$$\rho(\lambda,x,y).$$

Measurement independence is precisely the factorization

$$\rho(\lambda,x,y)=\rho(\lambda)\rho(x,y).$$

The proposed framework allows the more general case

$$\rho(\lambda,x,y) \neq \rho(\lambda)\rho(x,y).$$

Equivalently, provided the relevant probabilities are nonzero, one may express the dependence in either conditional direction:

$$P(\lambda|x,y) \neq P(\lambda)$$

or

$$P(x,y|\lambda) \neq P(x,y).$$

These are mathematically related descriptions of the same statistical dependence. Neither form, by itself, says which variable causally determines the other.

A.25 The Question of "Who Chose the Settings?"

The obvious objection is immediate: if λ is correlated with Alice's and Bob's settings, does that mean Alice and Bob were not free to choose their settings?

The answer requires care because the word *free* can refer to different concepts.

Bell's mathematical assumption is statistical independence. It says that the hidden variable and the measurement settings are statistically independent. It does not contain a complete philosophical theory of human freedom.

Conversely, rejecting statistical independence does not by itself specify a philosophical account of free will. It says only that the complete physical variables entering the experiment cannot be factorized in the way assumed by the Bell derivation.

This distinction will become the central subject of the next section. The fact that the mathematical proposal raises a question about free choice is not an accidental complication. It is the natural conceptual consequence of taking measurement dependence seriously.

A.26 Is the Correlation "Conspiratorial"?

A common objection to measurement-dependent models is that the required correlation appears extraordinarily coordinated. If the measurement settings are selected by complicated physical processes—perhaps involving random-number generators, human decisions, astronomical observations, or other distant events—why should the hidden variable associated with the source be correlated with those choices?

Calling the correlation "conspiratorial," however, is already an interpretive judgment. The mathematical statement itself is simply

$$P(\lambda|x,y) \neq P(\lambda).$$

Whether that dependence represents an implausible conspiracy, a global boundary condition, a retrocausal mechanism, a superdeterministic constraint, or some other physical principle cannot be determined from the equation alone.

The Block Universe perspective motivates a different possibility: perhaps the apparently independent events are not fundamentally independent because the complete physical history is selected as a globally consistent four-dimensional structure.

Whether such an account can be made physically natural rather than merely logically possible is an open question.

A.27 Relation to Retrocausal and Superdeterministic Terminology

The framework developed here lies close to ideas that are sometimes described in the foundations of physics literature as *retrocausal*, *measurement-dependent*, or *superdeterministic*. These labels overlap but are not identical.

A retrocausal model typically introduces a causal influence from a later boundary condition or measurement choice into an earlier physical variable. A superdeterministic model generally permits correlations between hidden variables and measurement settings such that the usual statistical independence assumption fails. A measurement-dependent model is the broader mathematical category in which

$$P(\lambda|a,b) \neq P(\lambda).$$

The present manuscript does not need to commit at this stage to any one of these labels. Its central claim is narrower: the Block Universe makes it conceptually possible to regard measurement dependence as a property of a complete spacetime history rather than automatically as a signal traveling backward in time.

A.28 The Experimental Achievement Remains Intact

Nothing in this proposal changes the empirical status of Bell experiments.

The experiments still demonstrate correlations that violate Bell inequalities. Quantum mechanics still predicts those correlations. The explicit measurement-dependent model reproduces the singlet probabilities.

The proposed reinterpretation therefore does not dispute the experimental result. It changes the question asked of the result.

Instead of asking:

"Which nonlocal influence produces the Bell correlation?"

the proposal asks:

"Can the Bell correlation be a constraint on the joint structure of a complete local physical history?"

The second question is the one that leads directly from quantum foundations back to the Block Universe.

A.29 The Remaining Physical Questions

The explicit construction establishes an important mathematical possibility, but several major questions remain unanswered.

- **What physical principle determines the joint distribution?** Why should complete histories be weighted in precisely the way required by quantum mechanics?
- **How is the distribution embedded in spacetime?** A complete theory must specify what physical object or law carries the relevant correlations.
- **Can the full Born rule be recovered?** Reproducing the singlet state is not enough to establish a general theory of quantum mechanics.
- **Can more complicated entangled states be reproduced?** A viable theory must extend beyond the simplest two-particle example.
- **Can relativistic covariance be maintained?** The proposal must not depend on an arbitrary preferred frame.
- **Can the apparent freedom of experimental choices be explained consistently?** This is the conceptual problem that naturally leads into Section 5.

These questions prevent the present argument from being presented as a finished alternative to quantum mechanics. What has been established is a coherent route by which one of Bell's assumptions can be relaxed while preserving local response functions and reproducing the singlet statistics.

A.30 The Proposed Physical Ingredients

The framework developed in this section can now be summarized by its proposed ingredients:

1. Four-dimensional spacetime described as a Block Universe.
2. Local response functions $A=A(x,\lambda)$ and $B=B(y,\lambda)$.
3. Quantum probabilities retained as genuine statistical structure.
4. Hidden variables associated with individual physical histories.
5. Measurement independence relaxed: $\rho(\lambda|x,y)$ need not equal $\rho(\lambda)$.
6. Exact quantum correlations reproduced by the joint distribution.
7. No operational superluminal signaling.

The central conceptual shift is from a picture in which each event is generated independently as time progresses to a picture in which the complete physical history is constrained as a whole.

A.31 From Local Dynamics to Global Consistency

This shift does not necessarily require abandoning local physics.

A local dynamical law specifies how physical quantities behave in a neighborhood of spacetime. A global consistency condition can additionally constrain which complete solutions of those local laws are physically realized.

The distinction can be expressed schematically:

Local dynamics → what relationships are permitted locally

Global consistency → which complete spacetime histories are realized

The proposal is that Bell correlations may belong partly to the second category. The local response of each measurement device remains local, while the complete history in which the device operates may be globally constrained.

This is precisely the kind of distinction that the Block Universe makes conceptually natural: the universe is not fundamentally a sequence of independent three-dimensional states being created one after another. It is a four-dimensional physical structure whose different regions must jointly satisfy the laws and boundary conditions governing the whole.